

Deep Learning for Video Anomaly Detection: A Review

Peng Wu^{ID}, Chengyu Pan, Yuting Yan, Guansong Pang^{ID}, *Member, IEEE*, Qingsen Yan^{ID}, *Member, IEEE*, Peng Wang^{ID}, *Member, IEEE*, and Yanning Zhang^{ID}, *Fellow, IEEE*

Abstract—Video anomaly detection (VAD) aims to discover behaviors or events deviating from the normality in videos. As a long-standing task in the field of computer vision, VAD has witnessed much good progress. In the era of deep learning, with the explosion of architectures of continuously growing capability and capacity, a great variety of deep learning-based methods are constantly emerging for the VAD task, greatly improving the generalization ability of detection algorithms and broadening the application scenarios. Therefore, such a multitude of methods and a large body of literature make a comprehensive survey a pressing necessity. In this article, we present an extensive and comprehensive research review, covering the spectrum of five different categories, namely, semi-supervised, weakly supervised, fully supervised, unsupervised, and open-set supervised VAD, and we also delve into the latest VAD works based on pretrained large models and open-world learning, remedying the limitations of past reviews in terms of only focusing on semi-supervised VAD and small model-based methods. For the VAD task with different levels of supervision, we construct a well-organized taxonomy, profoundly discuss the characteristics of different types of methods, and show their performance comparisons. In addition, this review involves the public datasets, open-source codes, and evaluation metrics covering all the aforementioned VAD tasks. Finally, we provide several important research directions for the VAD community. Additional details of the survey are available on the project homepage: <https://github.com/RocNg/DeepVAD>

Index Terms—Anomaly detection, deep learning, large models, video anomaly detection (VAD), video understanding.

I. INTRODUCTION

ANOMALY represents something that deviates from what is standard, normal, or expected. There are myriads of normalities, and anomalies are considerably scarce. However, when anomalies do appear, they often have a negative impact. Anomaly detection aims to discover these rare anomalies built

on top of machine learning, thereby reducing the cost of manual judgment. Anomaly detection has widespread application across various fields [1], such as financial fraud detection, network intrusion detection, industrial defect detection [2], and human violence detection. Among these, video anomaly detection (VAD) occupies an important place, in which anomaly indicates the abnormal events in the temporal or spatial dimensions. VAD not only plays a vital role in many fields, such as intelligent security and autonomous driving [3]. Owing to its significant potential for applications across different fields, VAD has attracted considerable attention from both industry and academia.

In the pre-deep learning era, the routine way is to separate feature extraction and classifier design, which forms a two-stage process, and then combine them together during the inference stage. First, there is a feature extraction process to convert the original high-dimensional raw videos into compact handcrafted features based on prior knowledge of experts. For example, commonly used features mainly include trajectory [4], histogram of flow (HOF) [5], 3-D gradient [6], mixture of dynamic textures (MDT) [7], and so on. Second, these expressed features are fed into the traditional machine learning algorithms for abnormal classification learning, where well-known algorithms include but are not limited to sparse reconstruction [5], [8], K -nearest neighbor (KNN) [9], and MDT [7]. Although handcrafted features lack robustness and are difficult to use for capturing effective behavior expressions in the face of complex scenarios and optimizing traditional algorithms is challenging, these pioneering works deeply enlighten subsequent deep learning-based works.

The rise of deep learning has made traditional machine learning algorithms fall out of favor over the last decade. With the rapid development of computer hardware and the massive data in the Internet era, we have witnessed great progress in developing deep learning-based methods for VAD in recent years: for example, ConvAE [10], the first work using deep autoencoders based on convolutional neural networks (CNNs) for capturing the regularities in videos; FuturePred [11], the first work making use of U-Net for forecasting future anomalies; and DeepMIL [12], the first endeavor exploring deep multiple instance learning (MIL) framework for real-world anomalies. In order to more intuitively manifest the research enthusiasm for the VAD task in the era of deep learning, we conduct a statistical survey on the number of publications related to VAD over the past decade (the era driven by the rise of deep learning-based methods) through Google Scholar and IEEE Xplore. We select five related topics, i.e., VAD, abnormal event detection, abnormal behavior detection, anomalous event detection, and anomalous behavior detection, and showcase the publication statistics in Fig. 1. It is not hard to see that the number of related publications counted

Received 20 April 2025; revised 21 September 2025; accepted 19 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62306240 and Grant U23B2013 and in part by China Postdoctoral Science Foundation under Grant 2023TQ0272. The work of Guansong Pang was supported in part by the Agency for Science, Technology and Research (A*STAR) under its MTC Young Individual Research Grant (YIRG) Grant M24N8c0103; in part by the Ministry of Education, Singapore, through its Tier-1 Academic Research Fund under Grant 24-SIS-SMU-008; and in part by the Lee Kong Chian Fellowship under Grant T050273. (Chengyu Pan and Yuting Yan contributed equally to this work.) (Corresponding author: Guansong Pang.)

Peng Wu, Chengyu Pan, Yuting Yan, Qingsen Yan, Peng Wang, and Yanning Zhang are with the School of Computer Science, Northwestern Polytechnical University, Xi'an, Shaanxi 710129, China (e-mail: xdwupeng@gmail.com; pancy5519@gmail.com; xgdyanyuting@gmail.com; qingsenyan@gmail.com; peng.wang@nwpu.edu.cn; ynzhang@nwpu.edu.cn).

Guansong Pang is with the School of Computing and Information Systems, Singapore Management University, Singapore 188065 (e-mail: pangguansong@gmail.com).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TNNLS.2025.3647892>, provided by the authors.

Digital Object Identifier 10.1109/TNNLS.2025.3647892

TABLE I
ANALYSIS AND COMPARISON OF RELATED REVIEWS

Reference	Year	Main Focus	Main Categorization	Research Topics						
				UVAD	WVAD	SVAD	FVAD	OVAD	LVAD	IVAD
Ramachandra et al. [13]	2020	Semi-supervised single-scene VAD	Methodology	✓	✓	✓	✓	✓	✓	✓
Santhosh et al. [14]	2020	VAD applied on road traffic	Methodology	✓	✓	✓	✓	✓	✓	✓
Nayak et al. [15]	2021	Deep learning driven semi-supervised VAD	Methodology	✓	✓	✓	✓	✓	✓	✓
Tran et al. [16]	2022	Semi&weakly supervised VAD	Architecture	✓	✓	✓	✓	✓	✓	✓
Chandrakala et al. [17]	2023	Deep model-based one&two-class VAD	Methodology&Architecture	✓	✓	✓	✓	✓	✓	✓
Liu et al. [18]	2023	Deep models for semi&weakly supervised VAD	Model Input	✓	✓	✓	✓	✓	✓	✓
NSVAD [19]	2025	Networking Systems VAD	Methodology&Network	✓	✓	✓	✓	✓	✓	✓
P2VAD [20]	2025	Privacy-Preserving VAD	Methodology&System	✓	✓	✓	✓	✓	✓	✓
Our survey	2025	Comprehensive VAD taxonomy and deep models	Methodology, Architecture, Refinement Model Input, Model Output	✓	✓	✓	✓	✓	✓	✓

UVAD: Unsupervised VAD, WVAD: Weakly supervised VAD, SVAD: Semi-supervised VAD, FVAD: Fully supervised VAD, OVAD: Open-set supervised VAD, LVAD: Large-model based VAD, IVAD: Interpretable VAD

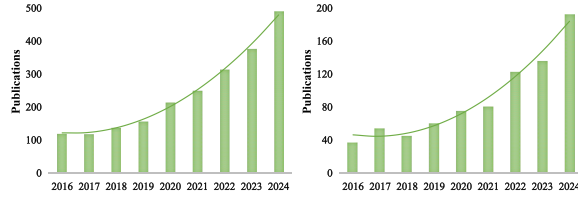


Fig. 1. Publications on VAD. Left: Google Scholar. Right: IEEE Xplore.

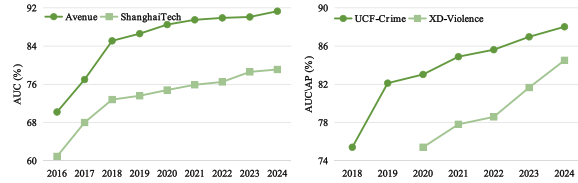


Fig. 2. Performance development for semi-supervised/weakly supervised VAD tasks.

from both sources exhibits a steady and rapid growth trend, demonstrating that VAD has garnered widespread attention. Moreover, we also demonstrate the detection performance trends of annual state-of-the-art methods on commonly used datasets under two common supervised manners and present performance trends in Fig. 2. The detection performance shows a steady upward trend across all datasets, without displaying any performance bottleneck. This indicates the evolving capability of deep learning methods under developing architectures and also showcases the ongoing exploration enthusiasm of deep learning methods for the VAD task.

The above statistics clearly demonstrate that deep learning-driven VAD is a hot area of current research. Therefore, there is an urgent necessity for a systematic taxonomy and comprehensive summary of existing works, to facilitate newcomers as a guide and provide references for existing researchers. Based on this, we first collect some high-profile reviews on VAD in the past few years, which are shown in Table I. Ramachandra et al. [13] mainly focused on semi-supervised VAD in the single scenario, lacking in discussions of cross-scenes. Santhosh et al. [14] reviewed VAD methods focusing on entities in road traffic scenarios. Their reviews lack sufficient in-depth analysis and center on pre-2020 methodologies, resulting in the neglect of recent advances. Nayak et al. [15] comprehensively surveyed on deep learning-based methods for semi-supervised VAD but did not take into account weakly supervised VAD methods. The follow-up work of Tran et al. [16] introduced a review of the emerging weakly supervised VAD, but the focus is not only on videos but also on image anomaly detection, resulting in a lack of systematic organization of the VAD task. Both Chandrakala et al. [17]

and Liu et al. [18] constructed an organized taxonomy covering a variety of VAD tasks, e.g., unsupervised VAD, semi-supervised VAD, weakly supervised VAD, and fully supervised VAD, and also surveyed deep learning-based methods for most supervised VAD tasks. However, they restrict their scope to the conventional closed-set scenario and fail to cover the latest research in the open-set supervised VAD, without introducing a brand-new pipeline based on pretrained large models and interpretable learning. More recently, Liu et al. [19], [20] presented two surveys on VAD: NSVAD and P2VAD, which mainly focus on networking system deployment and privacy-preserving techniques, respectively. While these works provide valuable insights from specific perspectives, they are limited to addressing one specific challenge in VAD, offering less comprehensive analysis of the VAD task itself.

To address this gap comprehensively, we present a thorough survey of VAD works in the deep learning era. Our survey covers several key aspects to provide a comprehensive analysis of VAD studies. To be specific, we perform an in-depth investigation into the development trends of VAD task in the era of deep learning and then propose a unified framework that integrates different VAD tasks together, filling the gaps in the existing reviews in terms of taxonomy. We then collect the most comprehensive open sources, including benchmark datasets, evaluation metrics, open-source codes, and performance comparisons, to help researchers in this field avoid detours and improve efficiency. Furthermore, we systematically categorize various VAD tasks, dividing existing works into different categories and establishing a clear and structured taxonomy system to provide a coherent and organized overview of various VAD paradigms. In addition to this taxonomy, we conduct a comprehensive analysis of each paradigm. Furthermore, throughout this survey, we spotlight influential works that have significantly contributed to the research advancement.

The main contributions of this survey are summarized in the following three aspects.

- 1) We provide a comprehensive review of VAD, covering five tasks based on different supervision signals, i.e., semi-supervised VAD, weakly supervised VAD, fully supervised VAD, unsupervised VAD, and open-set supervised VAD. The research focus has expanded from traditional single-task semi-supervised VAD to a broader range of multiple VAD tasks.
- 2) Staying abreast of the research trends, we review the latest studies on open-set supervised VAD. Moreover, we also revisit the most recent VAD methods based on pretrained large models and interpretable learning. The emergence of these methods elevates both the

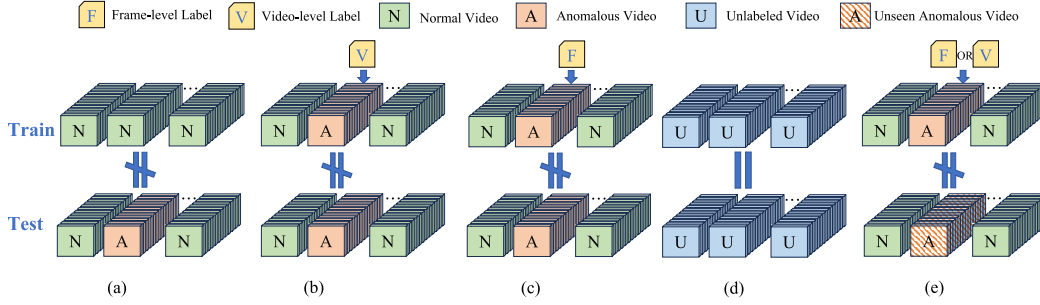


Fig. 3. Comparisons of five supervised VAD tasks. (a) Semi-supervised VAD. (b) Weakly supervised VAD. (c) Fully supervised VAD. (d) Unsupervised VAD. (e) Open-set supervised VAD.

TABLE II
COMPARISON OF FIVE DIFFERENT VAD TASKS

Tasks	Characteristics	Advantages	Disadvantages	Scenarios&Applications
SVAD	Only normal samples are available for model training	Efficient sample collection	High false positives	Airports and campuses: abandoned luggage detection
WVAD	Normal & abnormal samples are provided with coarse labels	Low labeling costs	Suboptimal predictions	Dataset expansion: shoplifting detection
FVAD	Anomaly samples are precisely annotated for model training	High detection accuracy	Intensive manual annotation	Fixed and high-security areas: fare evasion
UVAD	Training samples are completely unlabeled	No annotation burden	Poor detection performance	Intelligent traffic monitoring: wrong-way driving
OVAD	Detect unseen anomalies after training on known anomalies	High practical value	Poor model robustness	Dynamic urban environments: smart city alerts

performance and application prospects of VAD. To our knowledge, this is the first comprehensive survey of open-set supervised VAD, pretrained large model-based VAD, and interpretable VAD methods.

- For different tasks, we systematically review existing deep learning-based methods, and more importantly, we introduce a unified taxonomy framework categorizing the methods from various VAD paradigms based on various aspects, including model input, architecture, methodology, model refinement, and output. This meticulous scientific taxonomy enables a comprehensive understanding of the field.

The rest of this survey is structured as follows. Section II introduces an overview of the research background of VAD, e.g., the definition of different VAD tasks, benchmark datasets, and evaluation metrics. Sections III–VII present the semi-supervised, weakly supervised, fully supervised, unsupervised, and open-set supervised VAD methods, respectively. Section VIII provides a perspective on future development trends of VAD. Section IX draws meaningful conclusions for this survey.

II. BACKGROUND

A. Notation and Taxonomy

We demonstrate five different VAD tasks in Fig. 3 and also clearly showcase characteristics and comparisons of these tasks in Table II. The general VAD problem is presented as follows. Suppose that we are given a set of training samples $\mathcal{X} = \{x_i\}_{i=1}^{N+A}$ and corresponding labels $\mathcal{Y}$, where $\mathcal{X}_n = \{x_i\}_{i=1}^N$ is the set of normal samples and $\mathcal{X}_a = \{x_i\}_{i=N+1}^{N+A}$ is the set of abnormal samples. Each sample x_i is accompanied by a corresponding supervision label y_i in $\mathcal{Y}$. During the training phase, the detection model $\Phi(\theta)$ takes $\mathcal{X}$ as input and generates anomaly predictions; it is then optimized according to the following objective:

$$l = \mathcal{L}(\Phi(\theta, \mathcal{X}), \mathcal{X} \text{ or } \mathcal{Y}) \quad (1)$$

where $\mathcal{L}(\cdot)$ is employed to quantify the discrepancy between the predictions and the ground-truth labels or original samples. During inference, the detection model is expected to locate the abnormal behaviors or events in videos based on the generated anomaly predictions. Depending on the input to $\mathcal{L}$, VAD can be categorized into one of the following five tasks.

Semi-supervised VAD assumes that only normal samples are available during the training stage, that is, $\mathcal{X}_a = \emptyset$. This task aims to learn the normal patterns based on the training samples and consider the test samples that fall outside the learned patterns as anomalies. For *pros and cons*, only normal samples are required for training, and hence, there is no need to painstakingly collect scarce abnormal samples. However, any unseen test sample may be recognized as abnormal, leading to a higher false positive rate.

Weakly supervised VAD has more sufficient training samples and supervision signals than semi-supervised VAD. Both normal and abnormal samples are provided during the training stage, but the precise location stamps of anomalies in these untrimmed videos are unknown. In other words, only coarse video-level labels are available (i.e., inexact supervision). Formally, $\mathcal{Y} = \{0, 1\}_{i=1}^{N+A}$, where $y_i = 0$ indicates that x_i is normal and $y_i = 1$ indicates that x_i is abnormal. For *pros and cons*, compared to fully supervised annotations, it can significantly reduce labeling costs. However, it places higher demands on algorithm design and may lead to arbitrary or suboptimal predictions.

Fully supervised VAD, as its name implies, comprises the complete supervision signals, meaning that each abnormal sample has precise annotations of anomalies. This task can be viewed as a standard video or frame classification problem. Due to the scarcity of abnormal behaviors and intensive manual labeling in reality, there has been little research on the fully supervised VAD task. It is noteworthy that video violence detection can be regarded as a fully supervised VAD; hence, we denote violence detection as a fully supervised VAD task. Formally, each video x_i in $\mathcal{X}_a$ is accompanied by a corresponding supervision label $y_i = \{(t_j^s, t_j^e)\}_j^{U_i}$, where t_j^s and t_j^e

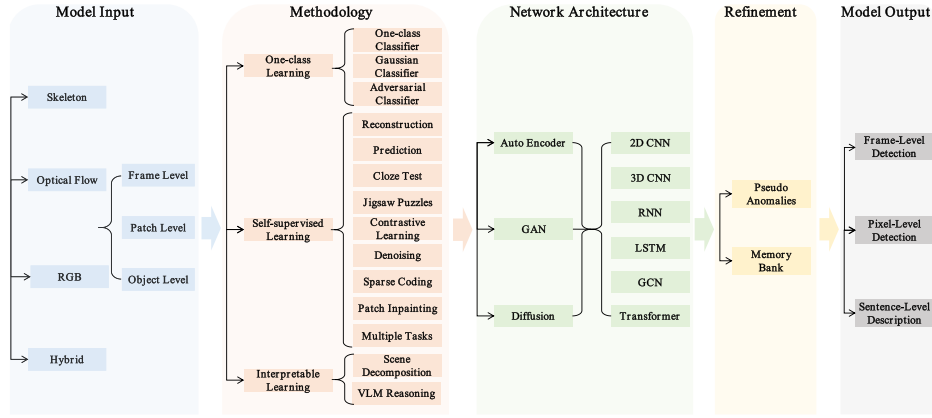


Fig. 4. Taxonomy of semi-supervised VAD. We provide a hierarchical taxonomy that organizes existing deep semi-supervised VAD models by model input, methodology, network architecture, refinement strategy, and model output into a systematic framework.

denote the start and end time of the j th violence event and U_i indicates the total number of anomalies present in the video. For *pros and cons*, in contrast to weakly supervised VAD, with full supervised information, the detection performance of the algorithms is remarkable. However, the corresponding drawback is the high requirement for intensive manual annotations.

Unsupervised VAD aims to discover anomalies directly from fully unlabeled videos in an unsupervised manner. Thus, unsupervised VAD no longer requires labeling normal and abnormal videos to build the training set. It can be expressed formally as follows: $\mathcal{X} = \mathcal{X}_{\text{test}}$ and $\mathcal{Y} = \emptyset$, in which $\mathcal{X}_{\text{test}}$ denotes the set of test samples. For *pros and cons*, no time-consuming effort is needed to collect training samples, avoiding the heavy labeling burden. Besides, this assumption also expands the application fields of VAD, implying that the detection system can continuously retrain without human intervention. Unfortunately, due to the lack of labels, the detection performance is relatively poor, leading to a higher rate of false positives and false negatives.

Open-set supervised VAD is devised to discover unseen anomalies that are not presented in the training set. Unlike semi-supervised VAD, open-set supervised VAD includes abnormal samples in the training set, which are referred to as seen anomalies. Specifically, for each x_i in $\mathcal{X}_a$, its corresponding label $y_i \in C_{\text{base}}$, here C_{base} represents the set of base (seen) anomaly categories and $C_{\text{base}} \subset \mathcal{C}$, with $\mathcal{C} = C_{\text{base}} \cup C_{\text{novel}}$. Here, C_{novel} and $\mathcal{C}$ represent the sets of novel anomaly categories unseen during training and all anomaly categories, respectively. Given a testing sample x_{test} , its label y_{test} may be either $\in C_{\text{base}}$ or $\in C_{\text{novel}}$. For *pros and cons*, compared to the two most common tasks, i.e., semi-supervised VAD and weakly supervised VAD, open-set supervised VAD not only reduces false positives but also avoids being limited to closed-set scenarios, thus demonstrating high practical value. However, due to the lack of prior knowledge, its robustness becomes a challenge. Therefore, it relies on learning specialized classifiers, loss functions, or generating unknown classes to detect unseen anomalies.

B. Datasets and Metrics

We showcase some representative VAD datasets with detailed information in the Supplementary Materials. We provide a more systematic overview of evaluation metrics across different levels, including frame-level metrics, pixel-level

metrics [21], as well as the latest sentence-level and reasoning-based metrics [22], [23], [24]. Due to space limitations, the detailed descriptions of more related benchmark datasets and evaluation metrics are showcased on the project website: <https://github.com/Roc-Ng/DeepVAD>. Moreover, work [25], which encompasses over 30 benchmarks for VAD, provides a comprehensive analysis of VAD-specific datasets.

III. SEMI-SUPERVISED VAD

Based on our in-depth investigation of past surveys, we found that previous surveys mostly lack a scientific taxonomy, in which many surveys simply categorize semi-supervised VAD works into different groups based on usage approaches, such as reconstruction-based, distance-based, and probability-based approaches, and some surveys classify works according to inputs, such as image-based, optical flow-based, and patch-based approaches. It is particularly apparent that existing classification reviews are relatively simplistic and superficial, thus making it challenging to cover all methods comprehensively and effectively. To address this issue, we establish a comprehensive taxonomy, encompassing model input, methodology, architecture, model refinement, and model output. The detailed illustration is presented in Fig. 4.

As aforementioned, only normal samples are available for training in the semi-supervised VAD task, rendering the supervised classification paradigm inapplicable. Common approaches involve leveraging the intrinsic information of the training samples to learn deep neural networks (DNNs) for solving a pretext task. For instance, normality reconstruction is a classic pretext task [10]. During this process, several critical aspects need consideration: selection of sample information (model input), design of pretext tasks (methodology), utilization of deep networks (network architecture), improvement of methods (refinement), and expression of anomaly results (model output). These key elements collectively contribute to the effectiveness of semi-supervised VAD solutions. In the following, we introduce existing deep learning-based VAD methods systematically according to the aforementioned taxonomy.

A. Model Input

Existing semi-supervised VAD methods typically use the raw video or its intuitive representations as the model input. Depending on the modality, these can be categorized as follows: RGB images, optical flow, skeleton, and hybrid

inputs, where the first three represent appearance, motion, and body posture.

1) *RGB*: RGB images are the most common input for conventional vision tasks driven by deep learning techniques, and this holds true for the VAD task as well. Unlike other modalities, RGB images do not require additional processing steps such as optical flow calculations or pose estimation algorithms. In the deep learning era, various deep models can be employed to extract compact and high-level visual features from these high-dimensional raw data. Utilizing these high-level features enables the design of more effective subsequent detection methods. Moreover, depending on the input size, RGB image-based input can be categorized into three principal groups: frame level, patch level, and object level.

Frame-level RGB captures the whole scene: background (often irrelevant) and foreground (likely anomaly regions). The conventional approach typically uses multiple consecutive video frames as a single input to capture temporal context within the video, as seen in methods like ConvAE [10], ConvLSTM-AE [26], and STAE [27]. On the other hand, several research studies focus on using single-frame RGB as input, aiming to detect anomalies at the spatial level, such as AnomalyGAN [28] and AMC [29].

Patch-level RGB input involves segmenting the frame-level RGB input spatially or spatiotemporally, which focuses on local regions, effectively separating the foreground from the background and differentiating between various individual entities. The primary advantage of patch-level input is its ability to significantly reduce the interference from the dominant background, which is usually less relevant to the anomalies. This segmentation helps in isolating areas that are more likely to contain anomalies, thereby enhancing the detection accuracy. For example, AMDN [30], [31], DeepOC [32], and Deep-cascade [33] took spatiotemporal patches as the input, and S^2 -VAE [34] and GM-VAE [35] only employed an image patch from a single video frame as the model input.

Object-level RGB input has emerged in recent years alongside the advancement of object detection approaches, which focuses exclusively on foreground objects. Compared to patch-level input, it entirely disregards background information and neglects to consider the relationship between objects and backgrounds. As a result, it demonstrates impressive performance in identifying anomalous events within complex scenes. Hinami et al. [36] first proposed an object-centric approach FRCN based on object inputs, a follow-up work ObjectAE [37] introduced to train object-centric autoencoders on detected objects, and subsequently, a series of works focusing on object-level inputs emerged, e.g., HF²-VAD [38], HSNBM [39], BDPN [40], ER-VAD [41], and HSC [42].

2) *Optical Flow*: A video is not merely a sequence of stacked RGB images; it encapsulates the temporal dimension and crucial temporal context. Therefore, extracting temporal context is vital for understanding video content, with motion information playing an irreplaceable role. Optical flow represents the motion information between consecutive video frames and is commonly used as a model input for the VAD task. It typically does not appear alone but is paired with the corresponding RGB image as input in the two-stream network. Therefore, it also encompasses frame [11], [43], [44], [45], [46], [47], patch [32], [48], [49], and object [38], [41], [47], [50] levels.

3) *Skeleton*: In recent years, with the remarkable success of deep learning technologies in the field of pose estimation, VAD methods based on skeleton input have emerged. Skeleton input solely focuses on the human body, making it more specialized than object-level RGB input. It demonstrates impressive performance in human-centric VAD, marking it as a significant research interest in recent years within the VAD domain. Morais et al. [51] first endeavored to learn the normal patterns of human movements using a dynamic skeleton, where the pose estimation is utilized to independently detect skeletons in each video frame. Then, GEPC [52], MTPP [53], NormalGraph [54], HSTGCNN [55], TSIF [56], STGCAE-LSTM [57], STGformer [58], STG-NF [59], MoPRL [60], MoCoDAD [61], and TrajREC [62] are specialized in human-related VAD with the skeleton input.

4) *Hybrid*: The hybrid input from different modalities often proves more advantageous for the VAD task compared to the unimodal input due to their complementary nature. In existing deep learning-driven VAD methods, hybrid input is a common practice. Typical hybrid input includes frame-level RGB combined with optical flow [11], patch-level RGB combined with optical flow [32], and object-level RGB combined with optical flow [38]. More recently, several studies have explored hybrid input based on RGB combined with skeleton [63].

B. Methodology

Through extensive research on existing works, we categorize three major approaches for learning the normal paradigm: self-supervised learning, one-class learning, and interpretable learning.

1) *Self-Supervised Learning*: Self-supervised learning primarily leverages auxiliary tasks (pretext tasks) to derive supervisory signals directly from unsupervised data. Essentially, self-supervised learning operates without external labeled data, as these labels are generated from the input data itself. For the semi-supervised VAD task, which lacks explicit supervisory signals, self-supervised learning naturally becomes essential for learning normal representations and constructing the normal paradigm/pattern based on these auxiliary tasks. Consequently, self-supervised learning-based methods consistently dominate the leading position in the semi-supervised VAD task. Throughout this process, a significant research focus and challenge lies in designing effective pretext tasks derived from the data itself. Here, we compile the common design principles of auxiliary tasks used in existing self-supervised learning-based methods.

Reconstruction is currently the most commonly used pretext task for self-supervised learning-based methods in the field of semi-supervised VAD [10], [26], [29], [34], [60], [64], [65], [66]. The main process involves inputting normal data into the network, performing encoding-decoding operations, and generating reconstructed data, encouraging the network to produce reconstructions that closely match the original input data. The objective can be expressed as

$$l_{\text{rec}} = \mathcal{L}(\Phi(\theta, x), x). \quad (2)$$

For convenience, in the following, unless otherwise specified, x represents normal data, which could be a normal video, a normal video frame, a normal feature, or similar. This loss quantifies reconstruction error, and the larger the reconstruction error, the higher the probability that the data is considered anomalous. However, due to the high capacity of DNNs, reconstruction-based methods cannot guarantee that

larger reconstruction errors for abnormal events necessarily happen.

Prediction fully leverages the temporal coherence inherent in videos, which is also a commonly used pretext task. This pretext task is based on the assumption that normal events are predictable, while abnormal ones are unpredictable. Specifically, the prediction pretext task takes historical data as input and, through encoding-decoding operations within the network, outputs the predicted data for the current moment. The network is compelled to make the predicted data similar to the actual current data. We define the optimization objective for prediction as

$$l_{\text{pre}} = \mathcal{L}(\Phi(\theta, \{I_{t-\Delta t}, \dots, I_{t-1}\}), I_t) \quad (3)$$

where I_t is the actual data at the current time step t and $I_{t-\Delta t:t-1}$ represents the historical data from time step $t - \Delta t$ to $t - 1$. FuturePred [11], as a future frame prediction framework, provided a new solution for VAD. Then, many researchers [38], [45], [58], [67], [68], [69], [70], [71], [72], [73] proposed other prediction-based methods. It alleviates, to some extent, the problem in reconstruction-based methods, where abnormal events can also be well reconstructed.

Visual cloze test is inspired by the cloze test in natural language processing [74], [75], [76]. It mainly involves training multiple DNNs to infer deliberately erased data from incomplete video sequences, where the prediction task can be considered a special case of visual cloze test task, i.e., the erased data happen to be the last frame in the video sequence. We define the objective function for completing erased data at the t th time stamp as

$$l_{\text{vct}}^{(t)} = \mathcal{L}(\Phi(\theta, \{I_1, \dots, I_{t-1}, I_{t+1}, \dots\}), I_t). \quad (4)$$

Similar to the prediction task, it also leverages the temporal relationships in the video, but the difference lies in this task can learn better high-level semantics and temporal context.

Jigsaw puzzles have recently been applied as a pretext task in semi-supervised VAD [77], [78], [79]. The main process involves creating jigsaw puzzles by performing temporal, spatial, or spatiotemporal shuffling, and then designing networks to predict the relative or absolute permutation in time, space, or both. The optimization function is as follows:

$$l_{\text{jig}} = \sum_i \mathcal{L}(t_i, \hat{t}_i) \quad (5)$$

where t_i and $\hat{t}_i$ denote the ground-truth and predicted positions of the i th data in the original sequence, respectively. Unlike the previous pretext tasks, which involve high-quality image generation, jigsaw puzzles are cast as a multilabel classification, enhancing computational efficiency and learning more contextual details.

Contrastive learning is a key approach in self-supervised learning, where the goal is to learn useful representations by distinguishing between similar and dissimilar pairs. For semi-supervised VAD, two samples are regarded as a positive pair if they originate from the same sample, and otherwise as a negative sample pair [80]. We show the contrastive loss as follows:

$$l_{\text{con}} = \sum_i -\log \frac{\exp(\text{sim}(x_i, x_i^+)/\tau)}{\sum_k \exp(\text{sim}(x_i, x_k^-)/\tau)} \quad (6)$$

where x_i and x_i^+ are the positive pair and x_i and x_k^- are the negative pair. $\text{sim}(\cdot, \cdot)$ is the similarity function (e.g., cosine

similarity). Wang et al. [81] introduced a cluster attention contrast framework for VAD, which is built on top of contrastive learning. During the inference stage, the highest similarity between the test sample and its variants is regarded as the regularity score. Lu et al. [82] further proposed a learnable locality-sensitive hashing with contrastive learning strategy for VAD.

Denoising [83], [84] is very similar to reconstruction, with the main difference being that noise η is added to the input data, and the network is encouraged to achieve a denoising effect for the reconstructed data. The benefit is that it can enhance the robustness of network for VAD. The optimization objective is expressed as

$$l_{\text{den}} = \mathcal{L}(\Phi(\theta, x + \eta), x). \quad (7)$$

Deep sparse coding is encouraged by the success of traditional sparse reconstruction-based VAD methods [5], and upgraded versions leverage DNNs for semi-supervised VAD. Unlike the aforementioned reconstruction or prediction tasks, sparse coding typically uses extracted high-level representations rather than raw video image data as input. By learning from a large amount of normal representations, a dictionary of normal patterns is constructed. The total objective is listed as

$$l_{\text{spa}} = \|x - Bz\|_2^2 + \|z\|_1. \quad (8)$$

Different normal events can be reconstructed through the dictionary B multiplied by the sparse coefficient z . For anomalies, it is hard to reconstruct them using the linear combination of elements from the normal dictionary with a sparse coefficient. To overcome the time-consuming inference and low-level handcrafted features of traditional sparse reconstruction-based methods, deep sparse coding-based methods have emerged [44], [85], [86], [87], simultaneously leveraging the powerful representation capabilities of DNNs and sparse representation techniques to improve detection performance and efficiency.

Patch inpainting involves the process of reconstructing missing or corrupted regions by inferring the missing parts from the available data. This technique mainly leverages the spatial and temporal context to predict the content of the missing regions, ensuring that the reconstructed regions blend seamlessly with the surrounding regions. The optimization objective for patch inpainting can be defined to minimize the difference between the original and the reconstructed patches,

$$l_{\text{pat}} = \mathcal{L}(\Phi(\theta, x \odot M), x \odot \bar{M}). \quad (9)$$

M denotes a mask, where the value of 0 in the mask indicates that the position needs to be inpainted, while a value of 1 indicates that the position does not need to be inpainted, and $\bar{M}$ is a reversed M . Different from prediction and visual cloze test, patch inpainting takes into account the spatial or spatiotemporal context. Zavrtanik et al. [88] regarded anomaly detection as a reconstruction-by-inpainting task, further randomly removed partial image regions, and reconstructed the image from partial inpaintings. Then, Ristea et al. [89] and Madan et al. [90] presented a novel self-supervised predictive architectural building block, a plug-and-play design that can easily be incorporated into various anomaly detection methods. More recently, a self-distilled masked autoencoder [91] was proposed to inpaint original frames.

Multiple task can ease the dilemma of the single pretext task, i.e., a single task may not be well aligned with the VAD task, thus leading to suboptimal performance. Recently, several

works attempted to train VAD models jointly on multiple pretext tasks. For example, various studies exploited different self-supervised task compositions, involving reconstruction and prediction [27], [39], [51], [92], prediction and denoising [93], [94], prediction and jigsaw puzzles [95], and prediction and contrastive learning [82]. A few works [78], [79], [96], [97] strove to develop more sophisticated multiple tasks from different perspectives.

2) *One-Class Learning*: One-class learning primarily focuses on samples from the normal class. Compared to self-supervised learning methods, it does not require the strenuous effort of designing feasible pretext tasks. It is generally divided into three categories: one-class classifier, Gaussian classifier, and adversarial classifier from discriminators of generative adversarial networks (GANs).

One-class classifier basically includes one-class support vector machine (OC-SVM) [98], support vector data description (SVDD) [99], and other extensions, e.g., basic/generalized one-class discriminative subspace classifier (BODS and GODS) [100]. Specifically, OC-SVM is modeled as an extension of the SVM objective by learning a max-margin hyperplane that separates the normal from the abnormal in a dataset, which is learned by minimizing the following objective:

$$\begin{aligned} \min_{w, b, \xi \geq 0} & \frac{1}{2} \|w\|_2^2 - b + C \sum \xi_i \\ \text{s.t. } & w^T x_i \geq b - \xi_i \quad \forall x_i \in \mathcal{X}_n \end{aligned} \quad (10)$$

where ξ_i is the nonnegative slack, w and b denote the hyperplane, and C is the slack penalty. AMDN [30] is a typical OC-SVM-based VAD method, which obtains low-dimensional representations through the autoencoder, and then uses OC-SVM to classify all normal representations. Another popular variant of one-class classifiers is (Deep) SVDD [101], [102] that, instead of modeling data to belong to an open half-space (as in OC-SVM), assumes the normal samples inhabit a bounded set, and the optimization seeks the centroid c of a hypersphere of minimum radius $R > 0$ that contains all normal samples. Mathematically, the objective reads

$$\begin{aligned} \min_{c, R, \xi \geq 0} & \frac{1}{2} R^2 + C \sum \xi_i \\ \text{s.t. } & \|x_i - c\|_2^2 \leq R^2 - \xi_i \quad \forall x_i \in \mathcal{X}_n \end{aligned} \quad (11)$$

where, as in OC-SVM, ξ_i models the slack. Based on this, Wu et al. [32] proposed an end-to-end deep one-class classifier, i.e., DeepOC, for VAD, avoiding the shortcomings of complicated two-stage training of AMDN.

Gaussian classifier-based methods [33], [35], [103] assume that, in practical applications, the data typically follow a Gaussian distribution. By using training samples, it learns the Gaussian distribution (mean μ and variance Σ) of the normal pattern. During the testing phase, samples that deviate significantly from the mean are considered anomalies. The abnormal score is presented as

$$p(x_i) = \frac{1}{(2\pi)^{\frac{k}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)\right). \quad (12)$$

Adversarial classifier uses adversarial training between the generator G and discriminator D to learn the distribution of normal samples. G is aware of normal data distribution, as the normal samples are accessible to it. Therefore, D explicitly

decides whether the output of G follows the normal distribution or not. Therefore, an adversarial classifier can be jointly learned by optimizing the following objective:

$$\begin{aligned} \min_G \max_D & \left(\mathbb{E}_{x_i \sim p_t} [\log(D(x_i))] \right. \\ & \left. + \mathbb{E}_{\tilde{x}_i \sim p_t + \mathcal{N}_\sigma} [\log(1 - D(G(\tilde{x}_i)))] \right) \end{aligned} \quad (13)$$

where x_i is drawn from a normal data distribution p_t and $\tilde{x}_i$ is the sample x_i with added noise, which is sampled from a normal distribution $\mathcal{N}_\sigma$. The final abnormal score of an input sample x is given as $D(G(x))$. For instance, Sabokrou et al. [104], [105], [106] developed a conventional adversarial network that contains two subnetworks, where the discriminator works as the one-class classifier, while the refiner supports it by enhancing the normal samples and distorting the anomalies. To mitigate the instability caused by adversarial training, Zaigham Zaheer et al. [107], [108] proposed stabilizing adversarial classifiers by transforming the role of discriminator to distinguish good and bad quality reconstructions as well as introducing pseudo-anomaly examples.

3) *Interpretable Learning*: While self-supervised and one-class learning methods perform well on standard VAD benchmarks, their reliance on end-to-end complex neural networks limits interpretability and generalization. To address this, explainable VAD aims not only to detect anomalies but also to provide reasons, which is essential for trust and transparency in real-world scenarios. The explainable VAD can be categorized into two types: scene decomposition-based and vision-language model (VLM) reasoning-based approaches.

Scene decomposition-based methods aim to disentangle the components of a scene to provide interpretable cues. For example, Hinami et al. [36] leveraged a multitask detector to extract generic knowledge of entities, actions, and attributes and then employed an environment-specific detector for anomaly recounting. Reiss and Hoshen [50] adopted both explicit (e.g., pose and velocity) and implicit semantic features for explainable decision-making. Doshi and Yilmaz [109] further utilized scene graphs to monitor individual and interactive behavior for contextual explanations. Singh et al. [110] developed a detector-independent model based on high-level visual features that can not only detect but also spatially localize anomalies with interpretable outputs.

VLM reasoning-based methods exploit the powerful inference capability of large language models (LLMs) and VLMs to provide human-understandable justifications. For example, Yang et al. [111] proposed the first rule-based reasoning framework for semi-supervised VAD by prompting LLMs to interpret abnormal events. VLAVAD [112] leveraged VLMs to enhance the interpretability of VAD through semantic feature reasoning instead of visual feature classification. More recently, SlowFastVAD [113] integrated a fast DNN-based detector with a slow VLM-based detector, by employing a retrieval-augmented generation (RAG)-driven reasoning process for a small number of ambiguous segments, it effectively balances inference speed and detection accuracy.

C. Network Architecture

1) *Autoencoder*: It consists of two vital structures, namely, encoder and decoder, in which the encoder compresses the input sample into a latent space representation, significantly reducing the dimensionality of the sample. The decoder restores the input sample from the latent space representation, increasing the dimensionality of the sample back to

the original input size. Due to its inherent image restoration and high-level representation extraction capabilities, an autoencoder is widely used in image restoration-based pretext tasks for self-supervised learning methods, such as reconstruction [10], prediction [27], and inpainting [91]. Moreover, an autoencoder is also used to extract features for one-class learning-based methods [32], where the extracted features are used for optimizing subsequent one-class classifiers. Autoencoder structures are highly flexible and can be based on various different base networks, such as 2-D CNN [10], 3-D CNN [27], recurrent neural network (RNN) [87], gated recurrent unit (GRU) [51], long short-term memory (LSTM) [26], [128], graph convolutional network (GCN) [54], [57], and Transformer [75], [91].

2) *Generative Adversarial Network*: GAN has been widely adopted in various applications, including VAD, due to its powerful generative capabilities. The key idea is to use the generator and discriminator to identify abnormal samples that deviate from the learned normal distribution. Specifically, GAN, like autoencoder, is mainly applied to image restoration-based pretext tasks [11], [129], [130], [131], [132], where the generator creates restored images and the discriminator is discarded once the training is finished. On the contrary, several one-class learning-based methods [104], [107] leverage the discriminator to assess the likelihood that a new sample is real (normal) or generated (anomalous). Low likelihood scores indicate anomalies, thereby achieving end-to-end one-class classifiers.

3) *Diffusion*: Diffusion models have achieved state-of-the-art performance across a wide range of generative tasks and have become a promising research topic. Unlike traditional generative models like GAN or autoencoder, diffusion models generate samples by reversing a diffusion process that gradually destroys sample structure. This reverse process reconstructs the sample from noise in a step-by-step manner, leading to high-quality results. Therefore, diffusion models also appear in the image restoration-based pretext tasks.

Flaborea et al. [61] and Yan et al. [124] introduced novel diffusion-based methods to predict the features using the skeleton and RGB frame, respectively, for VAD. Building on the preceding two methods, MA-PDM [133] introduced a unified framework that couples patch-level diffusion with dual appearance/motion conditioning, injecting fine-grained local information into the diffusion process to better capture anomalies.

D. Model Refinement

1) *Pseudo Anomalies*: In semi-supervised learning, there is usually a scarcity of real anomalous samples. To compensate for this lack, several research studies opt to generate pseudo anomalies. Current approaches include: *perturbing normal samples*, i.e., applying random perturbations to normal video samples, such as adding noise, shuffling frame sequences, or adding extra patches [134], [135], [136], [137]; *leveraging generative models*, i.e., using GAN or diffusion to generate samples similar to normal ones but with anomalous features [108], [138]; and *simulating specific anomalous behaviors*, i.e., manually introducing extra anomalous samples at the image-level or feature level [118], [125]. As a result, training with pseudo-anomalous samples allows the detection model to learn a variety of anomalous patterns and helps the model learn a broader range of anomalous features, improving its robustness and generalization abilities in real-world applications [139].

2) *Memory Bank*: Memory banks [120], [121], [122], [140], [141], [142] are used to store feature representations of normal video samples, which serve as a reference baseline and can be dynamically updated to adapt to new normal patterns, thereby enabling the model to better capture normal patterns and simultaneously improving the ability to adapt to changing environments. In specific implementations, memory banks can be combined with different network architectures, such as autoencoder-based reconstruction (or prediction) [114], [116], [117], [119] and contrastive learning [143].

It is worth noting that refinement strategies are designed as independent modules, which can be flexibly combined with different methodologies and network architectures rather than being restricted to specific counterparts. For instance, memory bank mechanisms can be integrated not only with sparse coding but also with prediction-based methods.

E. Model Output

1) *Frame-Level Detection*: In frame-level output, each frame of the video is classified as normal or abnormal. This output format provides an overall view of which frames in the video contain anomalies. Such output is simple and straightforward, easy to implement and understand, and particularly effective for detecting anomalies over a broad time range.

2) *Pixel-Level Detection*: In pixel-level output, not only is it identified which frames contain anomalies, but also which specific pixel regions within those frames are abnormal [144]. This output format provides more granular information about the anomalies. Pixel-level output offers precise locations and extents of anomalies, providing more detailed information for further analysis of the nature and cause of the anomalies.

3) *Sentence-Level Description*: Beyond identifying whether and where anomalies occur, sentence-level output provides natural language descriptions explaining the detected anomalies. Powered by LLMs and VLMs, this output offers reasoning about abnormal events, enabling the model to not only highlight anomalies but also to articulate why a specific event is considered anomalous [113].

F. Performance Comparison

Fig. 5 presents a concise chronology of semi-supervised VAD methods. Besides, Table III provides a performance summary observed in representative semi-supervised VAD methods.

IV. WEAKLY SUPERVISED VAD

Compared to semi-supervised VAD, weakly supervised VAD is a newer research direction, and therefore, existing reviews lack a comprehensive and in-depth introduction. As shown in Table I, Chandrakala et al. [17] and Liu et al. [18] mention the weakly supervised VAD task. However, the former only briefly describes several achievements from 2018 to 2020, while the latter, although encompassing recent works, lacks a scientific taxonomy and simply categorizes them into single-modal and multimodal based on the different modalities. Given this context, we survey related works from 2018 to the present, including the latest methods based on pretrained large models, and we classify existing works from four aspects: model input, methodology, refinement strategy, and model output. The taxonomy of weakly supervised VAD is illustrated in Fig. 6.

Compared to semi-supervised VAD, weakly supervised VAD explicitly defines anomalies during the training process,

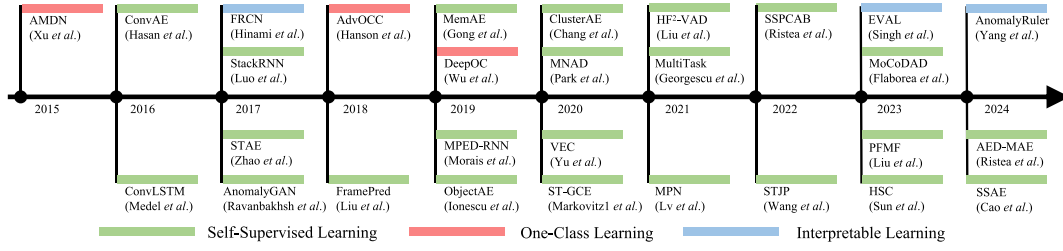


Fig. 5. Chronology of representative semi-supervised VAD research.

TABLE III
QUANTITATIVE PERFORMANCE COMPARISON OF SEMI-SUPERVISED METHODS ON PUBLIC DATASETS

Method	Publication	Methodology	Ped1	Ped2	Avenue	ShanghaiTech	UBnormal
			AUC (%)	AUC (%)	AUC (%)	AUC (%)	AUC (%)
AMDN [30]	BMVC 2015	One-class classifier	92.1	90.8	-	-	-
ConvAE [10]	CVPR 2016	Reconstruction	81.0	90.0	72.0	-	-
STAE [27]	ACMMM 2017	Hybrid	92.3	91.2	80.9	-	-
StackRNN [85]	ICCV 2017	Sparse coding	-	92.2	81.7	68.0	-
FuturePred [11]	CVPR 2018	Prediction	83.1	95.4	85.1	72.8	-
DeepOC [32]	TNNLS 2019	One-class classifier	83.5	96.9	86.6	-	-
MemAE [114]	ICCV 2019	Reconstruction	-	94.1	83.3	71.2	-
AnoPCN [92]	ACMMM 2019	Prediction	-	96.8	86.2	73.6	-
ObjectAE [37]	CVPR 2019	One-class classifier	-	97.8	90.4	84.9	-
BMAN [115]	TIP 2019	Prediction	-	96.6	90.0	76.2	-
sRNN-AE [87]	TPAMI 2019	Sparse coding	-	92.2	83.5	69.6	-
ClusterAE [64]	ECCV 2020	Reconstruction	-	96.5	86.0	73.3	-
MNAD [116]	CVPR 2020	Reconstruction	-	97.0	88.5	70.5	-
VEC [74]	ACMMM 2020	Cloze test	-	97.3	90.2	74.8	-
AMMC-Net [45]	AAAI 2021	Prediction	-	96.6	86.6	73.7	-
MPN [117]	CVPR 2021	Prediction	85.1	96.9	89.5	73.8	-
HF ² -VAD [38]	ICCV 2021	Hybrid	-	99.3	91.1	76.2	-
BAF [118]	TPAMI 2021	One-class classifier	-	98.7	92.3	82.7	59.3
Multitask [96]	CVPR 2021	Multiple tasks	-	99.8	92.8	90.2	-
F ² PN [43]	TPAMI 2022	Prediction	84.3	96.2	85.7	73.0	-
DLAN-AC [119]	ECCV 2022	Reconstruction	-	97.6	89.9	74.7	-
BDPN [40]	AAAI 2022	Prediction	-	98.3	90.3	78.1	-
CAFE [120]	ACMMM 2022	Prediction	-	98.4	92.6	77.0	-
STJP [77]	ECCV 2022	Jigsaw puzzle	-	99.0	92.2	84.3	56.4
MPT [78]	ICCV 2023	Multiple tasks	-	97.6	90.9	78.8	-
HSC [42]	CVPR 2023	Hybrid	-	98.1	93.7	83.4	-
LERF [121]	AAAI 2023	Prediction	-	99.4	91.5	78.6	-
DMAD [122]	CVPR 2023	Reconstruction	-	99.7	92.8	78.8	-
EVAL [110]	CVPR 2023	Interpretable learning	-	-	86.0	76.6	-
FBSC-AE [123]	CVPR 2023	Prediction	-	-	86.8	79.2	-
FPDM [124]	ICCV 2023	Prediction	-	-	90.1	78.6	62.7
PFMF [125]	CVPR 2023	Multiple tasks	-	-	93.6	85.0	-
STG-NF [59]	ICCV 2023	Gaussian classifier	-	-	-	85.9	71.8
AED-MAE [91]	CVPR 2024	Patch inpainting	-	95.4	91.3	79.1	58.5
SSMCTB [90]	TPAMI 2024	Patch inpainting	-	-	91.6	83.7	-
AnomalyRuler [111]	ECCV 2024	VLM Reasoning	-	97.9	89.7	85.2	71.9
SSAE [126]	TPAMI 2024	Prediction	-	-	90.2	80.5	-
EG-MPC [127]	CVPR 2025	Prediction	-	99.1	92.9	86.0	-

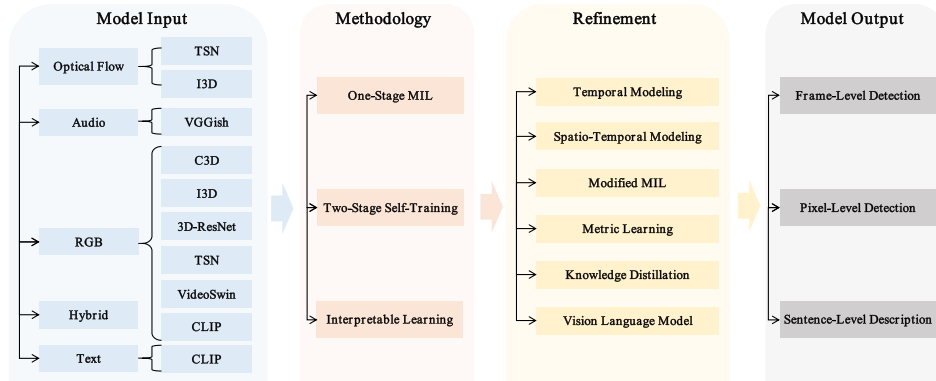


Fig. 6. Taxonomy of weakly supervised VAD. We provide a hierarchical taxonomy that organizes existing deep weakly supervised VAD models by model input, methodology, refinement strategy, and model output into a systematic framework.

giving the detection algorithm a clear direction. However, in contrast to fully supervised VAD, the coarse weak supervision

signals introduce uncertainty into the detection process. Most existing methods utilize the MIL mechanism to optimize the

model. This process can be viewed as selecting the hardest regions (video clips) that appear most abnormal from normal bags (normal videos) and the regions most likely to be abnormal from abnormal bags (abnormal videos). Then, the goal is to maximize the predicted confidence difference between them (with the confidence for the hardest normal regions approaching 0 and the confidence for the most abnormal regions approaching 1), which can be regarded as a binary classification optimization. By gradually mining all normal and abnormal regions based on their different characteristics, the anomaly confidence of abnormal regions increases, while that of normal regions decreases. Unfortunately, due to the lack of strong supervision signals, detection models tend to adopt speculative strategies during the optimization process, often resulting in ambiguous or inaccurate anomaly localization.

A. Model Input

Unlike semi-supervised VAD methods, recent weakly supervised VAD methods choose to use features extracted by pretrained models (e.g., CNN or Transformer-based representations of RGB, optical flow, or skeleton sequences) as input to the network backbone of anomaly detectors, instead of directly feeding the raw video frames. This approach alleviates the problem posed by the large scale of existing weakly supervised VAD datasets, the diverse and complex scenes, and the weak supervision signals. Using pretrained features as input allows the effective utilization of off-the-shelf models' learned knowledge of appearance and motion, significantly reduces the complexity of the detection model, and enables efficient training. However, it should be noted that in both semi-supervised and weakly supervised VAD, the original input source is still raw video; the main difference lies in whether feature extraction is separated as a preprocessing step or integrated into the end-to-end training.

1) *RGB*: RGB is the most common model input. The general approach divides a long video into multiple segments and uses pretrained visual models to extract global features from each segment. As deep models continue to evolve and improve, the visual models used have also been upgraded, progressing from the initial C3D [12], [145], [146] to I3D [147], [148], [149], [150], [151], 3D-ResNet [152], [153], TSN [154], [155], [156], and more recently, to the popular Swin Transformer [157], [158] and CLIP [159], [160]. This continuous upgrade in visual models has led to a gradual improvement in detection performance.

2) *Optical Flow*: Similar to RGB, the same approach is applied to optical flow input to obtain corresponding global features. However, due to the time-consuming nature of optical flow extraction, it is less commonly used in existing methods. Common pretrained models for optical flow include I3D [161] and TSN [155].

3) *Audio*: For multimodal datasets (e.g., XD-Violence) containing audio signals, audio also holds significant perceptual information. Unlike RGB images, audio is 1-D and is typically processed as follows: audios are resampled, computed the spectrograms, and created log mel spectrograms. Then, these features are framed into nonoverlapping examples. Finally, these examples are then fed into a pretrained audio model, such as VGGish [162], to extract features [163], [164].

4) *Text*: More recently, several researchers [165], [166], [167], [168] attempt to incorporate text descriptions related to videos to aid in VAD. These texts may be manually annotated or generated by large models. The text data are typically

converted into features using the text encoder and then fed into the subsequent detection network.

5) *Hybrid*: Common hybrid inputs include RGB combined with optical flow [161], RGB combined with audio [169], [170], [171], RGB combined with optical flow and audio [172], and, more recently, RGB combined with text [173].

B. Methodology

Based on existing methods, we divide them into three different groups: one-stage MIL, two-stage self-learning, and VLM-based interpretable learning.

1) *One-Stage MIL*: The one-stage MIL [12], [174], [175], [176] is the most commonly used approach for weakly supervised VAD. The basic idea is to first divide long videos into multiple segments and then use an MIL mechanism to select the most representative samples from these segments. This includes selecting hard examples from normal videos that look most like anomalies and the most likely abnormal examples from the abnormal videos. The model is then optimized by lowering the anomaly confidence of hard examples and increasing the confidence of the most likely abnormal examples. Ultimately, the confidence of model in predicting normal samples gradually decreases, while its confidence in predicting abnormal samples gradually increases, thereby achieving anomaly detection. The advantage of this method lies in its simplicity and ease of implementation. The MIL objective is showcased as

$$l_{\text{mil}} = \sum_i \max(0, 1 - \max(\Phi(\theta, x_i^a) + \max \Phi(\theta, x_i^n))) \quad (14)$$

where x^a and x^n denote an abnormal video and a normal video, respectively.

Additionally, TopK [148] extends MIL by selecting the top K segments with the highest prediction scores from each video, rather than just the highest scoring segment, for training. Therefore, MIL can be seen as a special case of TopK. For these TopK segments, compute their average prediction score as the predicted probability $\hat{y}$

$$\hat{y} = \sigma \left(\frac{1}{K} \sum_{i \in \text{topk}} \Phi(\theta, x_i) \right) \quad (15)$$

where σ is the sigmoid activation function. The cross-entropy loss between $\hat{y}$ and the label y is used to optimize the model

$$l_{\text{topk}} = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})) \quad (16)$$

The one-stage MIL mechanism leads to models that tend to focus only on the most significant anomalies while ignoring less obvious ones.

2) *Two-Stage Self-Training*: In contrast, improved two-stage self-learning is more complex but also more effective. This method employs a two-stage training process. First, a preliminary model is pretrained using the one-stage MIL. During this phase, the model learns the basic principles of VAD. Then, using the pretrained model as an initial parameter, a self-learning mechanism is introduced to adaptively train the model further, enhancing its ability to recognize anomalies. Specifically, during the self-learning phase, the predictions of model from the pretraining stage are used to automatically select high-confidence abnormal regions. These regions are then treated as pseudo-labeled data to retrain the model. The effectiveness of two-stage self-training mainly arises from the progressive refinement

of supervision signals. Initially, the model is trained with weak supervision, allowing it to capture coarse anomalous patterns. In the second stage, leveraging the predictions as pseudo-labels, the model receives more targeted guidance focusing on regions with high anomaly probability [158]. This iterative pseudo-labeling and retraining process helps the model to capture subtle and accurate abnormal events that are difficult to detect with only weak labels. As a result, two-stage self-training can significantly improve detection performance, especially in complex scenes with noisy or incomplete supervision. GCN-Clean [155], MIST [177], MSF [158], CUPL [178], and TPWNG [179] are typical two-stage self-training works.

The two-stage self-learning method based on improved MIL excels in weakly supervised VAD, but it also comes with some drawbacks, such as high computational complexity: the two-stage training process requires more computational resources and time. Both pretraining and self-learning phases involve multiple iterations of training, leading to high computational costs. Depending on initial model quality, the self-learning stage relies on the initial model generated during pretraining. If the quality of the initial model is poor, erroneous predictions may be treated as pseudo-labels, affecting subsequent training effectiveness.

3) *VLM-Based Interpretable Learning*: In recent years, VLMs have demonstrated remarkable potential and versatility across various vision tasks, giving rise to a novel paradigm, i.e., VLM-based interpretable learning, which enhances the interpretability of VAD from two perspectives.

- 1) *Category-Level Interpretability*: Unlike traditional methods that merely generate binary predictions (abnormal/normal), VLM-based approaches can explicitly predict the semantic category of anomalies by aligning visual content with language representations. However, most of these methods still rely on MIL or self-training schemes to guide anomaly localization and model optimization. For instance, VadCLIP [180] was the first to adopt an aligned paradigm based on CLIP to support both coarse-grained and fine-grained VAD. TPWNG [179] further extended VadCLIP by incorporating a two-stage self-training strategy within the CLIP-aligned framework.
- 2) *Semantic-Level Interpretability*: This approach focuses on leveraging LLMs and VLMs to generate natural language descriptions that explain detected anomalies. It typically involves four key steps.
 - a) Extracting high-level visual semantics from video frames using a visual encoder.
 - b) Mapping these features into the language space via VLMs to enable cross-modal alignment.
 - c) Constructing natural language prompts to guide reasoning within LLMs.
 - d) Generating explanatory descriptions of anomalies using LLMs.

These methods generally move beyond MIL and self-training strategies, instead relying on the intrinsic reasoning capabilities of VLMs to perform training-free anomaly detection and generate interpretable explanations. However, they often suffer from slower inference speeds and suboptimal detection performance in specific VAD scenarios. Representative works include HAWK [181], VERA [182], CUVA [23], VADor [183], and LAVAD [184]. Notably, although some methods, such as LAVAD [184], are essentially a training-

free unsupervised approach, we list them under the weakly supervised category for consistency in benchmark usage and comparative evaluation.

C. Refinement Strategy

Refinement strategies primarily focus on input features, method design, and other aspects to compensate for the shortcomings of weak supervision signals. We compile several commonly used refinement strategies and provide a detailed introduction in this section.

1) *Temporal Modeling*: Temporal modeling is essential for capturing the critical context information in videos. Unlike actions, anomalous events are complex combinations of scenes, entities, actions, and other elements, which require rich contextual information for accurate reasoning. Existing temporal modeling methods can be broadly categorized into local relationship modeling and global relationship modeling. Local modeling is typically used for online detection [186], whereas global modeling is mainly used for offline detection [185]. Techniques such as temporal convolutional networks [148], [194], dilated convolution [185], GCN [188], [195], conditional random field [196], and Transformers [158], [191], [197], [198], [199] are frequently employed to capture these temporal relationships effectively.

2) *Spatiotemporal Modeling*: Furthermore, spatiotemporal modeling can simultaneously capture spatial relationships, highlighting anomalous spatial locations and effectively reducing noise from irrelevant backgrounds. This can be achieved by segmenting video frames into multiple patches or using existing object detectors to capture foreground objects. Then, methods like self-attention [46], [153], [187], [200] are used to learn the relationships between these patches or objects. Compared to temporal modeling, spatiotemporal modeling involves a higher computational load due to the increased number of entities being analyzed.

3) *MIL-Based Refinement*: The traditional MIL mechanism focuses only on the segments with the highest anomaly scores, which leads to a series of issues, such as ignoring event continuity, the fixed length K -value not adapting to different video scenarios, and bias toward abnormal snippets with simple contexts. Several advanced strategies [201], [202] aim to address the limitations. By incorporating unbiased MIL [190], prior information from text [167], [192], magnitude-level MIL [189], continuity-aware refinement [203], and adaptive K -values [204], the detection performance can be significantly improved.

4) *Feature Metric Learning*: While MIL-based classification ensures the interclass separability of features, this separability at the video level alone is insufficient for accurate anomaly detection. In contrast, enhancing the discriminative power of features through clustering similar features and isolating different ones should complement and even augment the separability achieved by MIL-based classification. Specifically, the basic principle of feature metric learning is to make similar features compact and different features distant in the feature space to improve discrimination. Several works [150], [165], [167], [186], [197], [205], [206] exploited feature metric learning to enhance the feature discrimination.

5) *Knowledge Distillation*: It aims to transfer knowledge from the enriched branch to the barren branch to alleviate the semantic gap, which is mainly applied for modality-missing [207] or modality-enhancing [171] scenarios.

TABLE IV
QUANTITATIVE PERFORMANCE COMPARISON OF WEAKLY SUPERVISED METHODS ON PUBLIC DATASETS

Method	Publication	Feature	UCF-Crime	XD-Violence	ShanghaiTech	TAD
			AUC (%)	AP (%)	AUC (%)	AUC (%)
DeepMIL [12]	CVPR 2018	C3D ^{RGB}	75.40	-	-	-
GCN-Clean [155]	CVPR 2019	TSN ^{RGB}	82.12	-	84.44	-
HLNet [148]	ECCV 2020	I3D ^{RGB}	82.44	75.41	-	-
CLAWS [146]	ECCV 2020	C3D ^{RGB}	83.03	-	89.67	-
MIST [177]	CVPR 2021	I3D ^{RGB}	82.30	-	94.83	-
RTFM [185]	ICCV 2021	I3D ^{RGB}	84.30	77.81	97.21	-
CTR [186]	TIP 2021	I3D ^{RGB}	84.89	75.90	97.48	-
MSL [158]	AAAI 2022	VideoSwin ^{RGB}	85.62	78.59	97.32	-
S3R [149]	ECCV 2022	I3D ^{RGB}	85.99	80.26	97.48	-
SSRL [187]	ECCV 2022	I3D ^{RGB}	87.43	-	97.98	-
CMRL [188]	CVPR 2023	I3D ^{RGB}	86.10	81.30	97.60	-
CUPL [178]	CVPR 2023	I3D ^{RGB}	86.22	81.43	-	91.66
MGFN [189]	AAAI 2023	VideoSwin ^{RGB}	86.67	80.11	-	-
UMIL [190]	CVPR 2023	CLIP-ViT	86.75	-	-	92.93
DMU [191]	AAAI 2023	I3D ^{RGB}	86.97	81.66	-	-
PE-MIL [192]	CVPR 2024	I3D ^{RGB}	86.83	88.05	98.35	-
TPWNG [179]	CVPR 2024	CLIP-ViT	87.79	83.68	-	-
VadCLIP [180]	AAAI 2024	CLIP-ViT	88.02	84.51	-	-
STPrompt [193]	ACMMM 2024	CLIP-ViT	88.08	-	97.81	-
PEL4VAD [165]	TIP 2024	I3D ^{RGB}	86.76	85.59	98.14	-

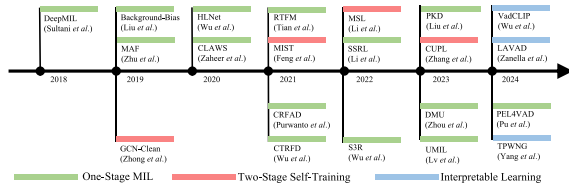


Fig. 7. Chronology for representative weakly supervised VAD research.

6) *Vision-Language Models*: VLMs enhance anomaly detection not only by incorporating language-informed visual representations, e.g., CLIP-TSA [208] and cross-modal semantic alignment, but also by LLMs to generate explanatory captions that improve detection performance, as demonstrated in TEVAD [166], UCA [173], and VAD-Instruct50k [22]. Moreover, they can directly exploit the prior knowledge encoded in LLMs and VLMs [184].

D. Model Output

Similar to semi-supervised VAD, the model output in weakly supervised VAD typically includes frame-level detection, pixel-level detection [209], [210], and sentence-level description.

E. Performance Comparison

As depicted in Fig. 7, several significant research works have emerged in the field's chronology. Moreover, we present an elaborate performance comparison of existing research, as detailed in Table IV.

V. FULLY SUPERVISED VAD

Fully supervised VAD refers to the task of detecting video anomalies under the condition that the dataset has detailed frame-level or video-level annotations. Here, we consider video violence detection as a fully supervised VAD task.

A. Approach Categorization

Video violence detection typically takes the appearance, motion, skeleton, audio, or a combination of these as the

input. It can be categorized based on the type of input into the following types.

Appearance input mainly consists of raw RGB images, directly showcasing the visual effect of video frames. This helps the model better understand anomalies that can be directly detected from a visual perspective. Many methods [211], [212], [220], [221] used RGB features extracted from raw images using pretrained models as the model input.

Motion input mainly includes optical flow, optical flow acceleration, and frame differences. These inputs directly showcase the motion state of objects, helping to identify anomalies from the motion perspective that might be difficult to detect visually. Dong et al. [211] and Peixoto et al. [222] used optical flow and optical flow acceleration as input, while Sudhakaran and Lanz [213] and Hanson et al. [214] employed frame differences as model input.

Skeleton input can intuitively display the pose state of humans, allowing the model to exclude background interference and focus on human actions. This enables more intuitive and vivid recognition of violent behavior. Su et al. [215] and Singh et al. [223] conducted violence detection by studying the interaction relationships between skeletal points.

Audio input can provide additional information to aid in identifying violent events [222]. This is because certain violent incidents inevitably involve changes in sound; such variations help us better detect violent events, especially when RGB images may not be effectively detected due to issues like occlusion.

Hybrid input combines the strengths of different modalities to better detect violent events. Cheng et al. [216] utilized RGB images and optical flow as input, while Shang et al. [224] combined RGB images with audio as input. Garcia-Cobo and SanMiguel [218] fed skeleton and frame differences into detection models.

B. Performance Comparison

We present the performance comparison of existing fully supervised VAD research in Table V.

VI. UNSUPERVISED VAD

Despite the great popularity of supervised VAD, supervised methods still have shortcomings in practical applications.

TABLE V
QUANTITATIVE PERFORMANCE COMPARISON OF FULLY SUPERVISED METHODS ON PUBLIC DATASETS

Method	Publication	Model Input	Hockey Fights	Violent-Flows	RWF-2000	Crowd Violence
			Accuracy (%)	Accuracy (%)	Accuracy (%)	Accuracy (%)
TS-LSTM [211]	PR 2016	RGB+Flow	93.9	-	-	-
FightNet [212]	JPCS 2017	RGB+Flow	97.0	-	-	-
ConvLSTM [213]	AVSS 2017	Frame Difference	97.1	94.6	-	-
BiConvLSTM [214]	ECCVW 2018	Frame Difference	98.1	96.3	-	-
SPIL [215]	ECCV 2020	Skeleton	96.8	-	89.3	94.5
FlowGatedNet [216]	ICPR 2020	RGB+Flow	98.0	-	87.3	88.9
X3D [217]	AVSS 2022	RGB	-	98.0	94.0	-
HSCD [218]	CVIU 2023	Skeleton+Frame Difference	94.5	-	90.3	94.3
RTPNet [219]	CVIU 2024	Skeleton+Frame	99.4	-	93.3	-

On the one hand, we cannot clearly define what constitutes normal behavior of real-life human activities in many cases, e.g., running in a sports ground is normal, but running in a library is forbidden. On the other hand, it is impractical to know every possible normal event in advance, especially for scientific research. Therefore, unsupervised VAD is of significant research value [235].

A. Approach Categorization

Through an in-depth investigation, we roughly classify the current unsupervised VAD methods into three categories: pseudo label, change detection, and others.

Pseudo-label-based paradigm is described as follows. Wang et al. [228] proposed a two-stage training approach where an autoencoder is first trained with an adaptive reconstruction loss threshold to estimate normal events from unlabeled videos. These estimated normal events are then used as pseudo-labels to train an OC-SVM, refining the normality model to exclude anomalies and improve detection performance. Pang et al. [229] introduced a self-training deep ordinal regression method, starting with initial detection using classical one-class algorithms to generate pseudo-labels for anomalous and normal frames. An end-to-end anomaly score learner is then trained iteratively using a self-training strategy that optimizes the detector with newly generated pseudo-labels. Zaheer et al. [236] proposed an unsupervised generative cooperative learning approach, leveraging the low-frequency nature of anomalies for cross-supervision between a generator and a discriminator, each learning from the pseudo-labels of the other. Al-lahham et al. [237] presented a coarse-to-fine pseudo-label generation framework using hierarchical divisive clustering for coarse pseudo-labels at the video level and statistical hypothesis testing for fine pseudo-labels at the segment level, training the anomaly detector with the obtained pseudo-labels.

Change detection-based paradigm can be summarized as follows. Del Giorno et al. [225] performed change detection in video frames using a simple logistic regression classifier to measure deviations between the data, making comparisons time-independent by randomly ordering frames. Ionescu et al. [226] proposed a change detection framework based on the unmasking technique, determining abnormal events by observing changes in classifier accuracy between consecutive events. Liu et al. [227] linked the heuristic unmasking procedure to multiple classifier two-sample tests in statistical machine learning, aiming to improve the unmasking method. Hu et al. [230] introduced a method based on masked autoencoders [238], where the rare and unusual nature of anomalies leads to poor prediction of changing events, allowing for anomaly detection and scoring in unlabeled videos.

Other paradigm includes the following methods. Li et al. [239] proposed a clustering technique, where an autoencoder is trained on a subset of normal data and iterates between hypothetical normal candidates based on clustering and representation learning. The reconstruction error is used as a scoring function to assess normality. Lin et al. [231] introduced a causal inference framework to reduce the effect of noisy pseudo-labeling, combining long-term temporal context with local image context for anomaly detection. Yu et al. [232] highlighted the effectiveness of deep reconstruction for unsupervised VAD, revealing a normality advantage where normal events have lower reconstruction loss. They integrated a novel self-paced refinement scheme into localization-based reconstruction for unsupervised VAD.

B. Performance Comparison

We present the performance comparison of existing unsupervised VAD research in Table VI.

VII. OPEN-SET SUPERVISED VAD

Open-set supervised VAD is a challenging task where the goal is to detect anomalous events in videos that are unseen during the training phase. Unlike traditional (closed-set) VAD, where the types of anomalies are known and well-defined, open-set VAD must handle unforeseen and unknown anomalies. This is crucial for real-world applications, as it is impractical to anticipate and annotate every possible anomaly during training. Therefore, research on open-set VAD has garnered significant attention. However, existing review works lack an investigation into open-set VAD. Based on this, we conduct an in-depth survey and make a systematic taxonomy of existing open-set VAD works. To our knowledge, this is the first review that includes a detailed introduction to open-set supervised VAD. In this section, we broadly categorize open-set supervised VAD into two types based on different research directions: open-set VAD and few-set VAD.

A. Open-Set VAD

Open-set VAD is a vital area of research that addresses the limitations of traditional anomaly detection methods by focusing on the detection of unknown anomalies. MLEP [240], as the first work on open-set supervised VAD, addressed the challenge of detecting anomalies in videos when only a few anomalous examples are available for training. It emphasizes learning a margin that separates normal and anomalous samples in the feature space. This helps in distinguishing anomalies even when only a few examples are available. The follow-up work [241] introduced a new benchmark, namely, UBnormal, specifically designed for open-set VAD. It aims to provide a comprehensive evaluation

TABLE VI
QUANTITATIVE PERFORMANCE COMPARISON OF UNSUPERVISED METHODS ON PUBLIC DATASETS

Method	Publication	Methodology	Avenue	Subway Exit	Ped1	Ped2	ShanghaiTech	UMN
			AUC (%)	AUC (%)	AUC (%)	AUC (%)	AUC (%)	AUC (%)
ADF [225]	ECCV 2016	Change Detection	78.3	82.4	-	-	-	91.0
Unmasking [226]	ICCV 2017	Change Detection	80.6	86.3	68.4	82.2	-	95.1
MC2ST [227]	BMVC 2018	Change Detection	84.4	93.1	71.8	87.5	-	-
DAW [228]	ACMMM 2018	Pseudo Label	85.3	84.5	77.8	96.4	-	-
STDOR [229]	CVPR 2020	Pseudo Label	-	92.7	71.7	83.2	-	97.4
TMAE [230]	ICME 2022	Change Detection	89.8	-	75.7	94.1	71.4	-
CIL [231]	AAAI 2022	Others	90.3	97.6	84.9	99.4	-	100
LBR-SPR [232]	CVPR 2022	Others	92.8	-	81.1	97.2	72.6	-
LANP [233]	ECCV 2024	Pseudo Label	-	-	-	-	86.5	-
FRD [234]	TMM 2024	Pseudo Label	91.2	-	-	-	80.1	-

framework for testing and comparing various VAD methods under open-set conditions. Zhu et al. [242] broke through the closed-set detection limitations and developed novel techniques that can generalize to previously unseen anomalies and effectively distinguish them from normal events. Specifically, a normalizing flow model is introduced to create pseudo-anomaly features. Recently, Wu et al. [160] extended open-set VAD to the more challenging open-vocabulary VAD, aiming to both detect and recognize anomaly categories. Centered around VLMs, this task is achieved by matching videos with corresponding textual labels. Additionally, large generative models and language models are utilized to generate pseudo-anomalous samples. There are other methods for the open-set setting, such as [243] and [244], but they are focused on image-level anomaly detection. Through various innovative approaches, such as margin learning, the development of benchmarks, generalization strategies, and the integration of language-vision models, researchers are pushing the boundaries of what is possible in VAD. These advancements are paving the way for more robust, flexible, and practical VAD systems suitable for a wide range of real-world applications.

B. Few-Shot VAD

The goal of few-shot VAD is to detect anomalies in a previously unseen scene with only a few frames. Compared to open-set VAD, the main difference lies in that a few real frames of unseen anomalies are provided. This task is first introduced by Lu et al. [245], and a meta-learning-based model is proposed to tackle this problem. During the test stage, the model needs to be fine-tuned via a few provided samples from the new scene. In order to avoid extra fine-tuning processes before deployment, Hu et al. [246] and Huang et al. [247] adopted the metric-based adaptive network and variational network, respectively, both leveraging a few normal samples as the reference during the test stage without any fine-tuning. Furthermore, Aich et al. [248] presented a novel zxVAD framework, a significant advancement by enabling anomaly detection across domains without requiring target domain adaptation. In this work, a novel untrained CNN-based anomaly synthesis module crafts pseudo-abnormal examples by adding foreign objects in normal video frames in a training-free manner. This contrasts with the above few-shot adaptive methods, which require a few labeled data from the target domain for fine-tuning. The former focuses on domain-invariant feature extraction and unsupervised learning, ensuring robustness and generalizability, while the latter relies on few-shot learning to adapt models to new domains with minimal labeled data. More recently, Jain et al. [249] proposed an extended version utilizing weakly

supervised learning to achieve cross-domain generalization. By incorporating external datasets from diverse domains, the model trained in a weakly supervised manner is enabled to generalize across domains.

VIII. FUTURE OPPORTUNITIES

A. Creating Comprehensive Benchmarks

The current VAD benchmarks have various limitations in terms of data size, modality, and capturing views. Thus, an important future direction is to extend benchmarks along these dimensions for providing more realistic VAD test platforms.

1) *Large-Scale*: Currently, in VAD, particularly in semi-supervised settings, the available datasets are of limited scale. Small-scale benchmarks tend to overestimate model accuracy, while performance in real-world scenarios remains significantly limited. Therefore, expanding the scale and diversity of VAD datasets is a critical direction for future research to ensure both robustness and real-world applicability.

2) *Multimodal*: Currently, there is limited research on multimodal VAD. Just as humans perceive the world through multiple senses, such as vision, hearing, and smell, effectively utilizing various modalities of information in the face of multisource heterogeneous data can enhance the performance of anomaly detection. For example, using audio information can better detect anomalies such as screams and panic, while using infrared information can identify abnormal situations in dark environments.

3) *Egocentric, Multiview, and 3-D*: Egocentric VAD involves using data captured from wearable devices or body-mounted cameras to simulate how individuals perceive their environment and identify abnormal events, such as detecting falls or aggressive behavior in real time. Creating multiview benchmarks that leverage data from viewpoints allows for a comprehensive environment analysis, enabling the detection of anomalies that may not be visible from a single perspective. The 3-D perspectives from depth information or point clouds can offer more detailed spatial information, enabling models to better understand the structure and context of the environment, which also brings multimodal signals.

However, deploying VAD in practice faces key challenges: large and diverse data complicates storage and annotation, real-time labeling is labor-intensive and constrained by ethical and legal issues, hardware limits and costs affect reliability, and dynamic environments demand advanced algorithms to integrate heterogeneous and 3-D sensor data.

B. Toward Open-World Task

The current research focuses on the closed-set VAD, which is restricted to detecting only those anomalies that are defined

and annotated during training. In applications like urban surveillance, the inability to adapt to unforeseen anomalies limits the practicality and effectiveness of closed-set VAD models. Therefore, moving toward the open-world VAD task, handling the uncertainty and variability of real-world situations, is a feasible future trend. To accomplish this task, several key approaches and their combination can be taken into account.

- 1) *Self-Supervised Learning*: Leveraging unlabeled data to learn discriminative representations that can distinguish between normal and abnormal events.
- 2) *Open-Vocabulary Learning*: Developing models that can adapt to new anomalies with large models [160], pseudo-anomaly synthesis, or minimal labeled examples.
- 3) *Incremental Learning*: continuously updating models with new data and anomaly types without forgetting previously learned information. Open-world VAD must handle concept drift, where the definition of normal and abnormal can change over time due to social, seasonal, or environmental factors. This requires models to be robust against distribution shifts and to learn continuously with minimal human intervention.

C. Embracing Pretrained Large Models

Pretrained large models have shown remarkable success in various computer vision tasks, and these models can be leveraged in VAD to enhance the understanding and detection of anomalies by integrating semantic context and improving feature representations. Here are several feasible directions.

- 1) *Feature Extraction*: Pretrained weights of large models, which have been trained on large-scale datasets, provide a strong foundation for feature extraction and reduce the need for extensive training from scratch [208].
- 2) *Semantic Understanding*: Language-vision models can be utilized to understand and incorporate contextual information from video scenes. For instance, text descriptions associated with video frames can provide additional context that helps in identifying anomalies. In the same way, leverage the language capabilities of these models to generate or understand textual descriptions of anomalies, aiding in both the detection and interpretation of the anomalies [22].
- 3) *Zero-Shot Learning*: It exploits the zero-shot learning capabilities of language-vision models to detect anomalies without requiring explicit examples during training. This is particularly useful in open-set VAD scenarios where new types of anomalies can occur [250]. The deployment of large pretrained models in real-time VAD faces challenges such as computational efficiency, high memory consumption, and the need for fast adaptation to environment-specific data.

D. Exploiting Interpretable VAD

Interpretable VAD focuses on creating models that not only detect anomalies but also provide understandable explanations for their predictions. This is crucial for gaining trust in the system, especially in high-stakes applications like surveillance, healthcare, and autonomous vehicles. Here are several feasible directions from three different layers of a VAD system.

- 1) *Input*: Instead of directly inputting raw video data into the model, leverage existing technologies to extract key information, such as foreground objects, position coordinates, motion trajectories, and crowd relationships.

- 2) *Algorithm*: Combining algorithms from different domains can be helpful for enhanced reasoning, including: knowledge graphs, i.e., utilize knowledge graphs to incorporate contextual information and relationships between entities; intent prediction, i.e., use intent prediction algorithms to anticipate future actions and detect deviations from expected behaviors [123], [126]; LLMs' reasoning, i.e., the textual descriptions of detected anomalies using large LLMs can also be used for the explanation. These descriptions can explain what the model perceives as abnormal and why [22].
- 3) *Output*: Various aspects such as the spatiotemporal changes and patterns in the video may be synthesized to explain anomalies [193]. In practical applications, decision transparency and user trust are crucial, especially for emergency-triggering anomalies. Generating accurate, human-understandable explanations is challenging in complex or crowded scenes, and interpretable models need to balance explanation detail with real-time detection to support timely interventions.

We provide further perspectives in the Supplementary Materials, including privacy-preserving VAD [205], [235], hallucination in VLMs, spatial-temporal reasoning [23], [24], and long-video understanding [22], to present a broader view of future directions.

IX. CONCLUSION

We present a comprehensive survey of VAD approaches in the deep learning era. Unlike previous reviews mainly focusing on semi-supervised VAD, we provide a taxonomy that systematically divides the existing works into five categories by their supervision signals, e.g., semi-supervised, weakly supervised, unsupervised, fully supervised, and open-set supervised VAD. For each category, we further refine the categories based on model differences, e.g., model input and output, methodology, refinement strategy, and architecture, and we demonstrate the performance comparison of various methods. Finally, we discuss several promising research directions for deep learning-based VAD in the future.

REFERENCES

- [1] G. Pang, C. Shen, L. Cao, and A. van den Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2021.
- [2] P. Bergmann, K. Batzner, M. Fauser, D. Sattlegger, and C. Steger, "The MVTEC anomaly detection dataset: A comprehensive real-world dataset for unsupervised anomaly detection," *Int. J. Comput. Vis.*, vol. 129, no. 4, pp. 1038–1059, Apr. 2021.
- [3] Y. Yao et al., "DoTA: Unsupervised detection of traffic anomaly in driving videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 444–459, Jan. 2023.
- [4] Z. Chen, Y. Tian, W. Zeng, and T. Huang, "Detecting abnormal behaviors in surveillance videos based on fuzzy clustering and multiple auto-encoders," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jun. 2015, pp. 1–6.
- [5] Y. Cong, J. Yuan, and J. Liu, "Sparse reconstruction cost for abnormal event detection," in *Proc. CVPR*, Jun. 2011, pp. 3449–3456.
- [6] C. Lu, J. Shi, and J. Jia, "Abnormal event detection at 150 FPS in MATLAB," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2720–2727.
- [7] W. Li, V. Mahadevan, and N. Vasconcelos, "Anomaly detection and localization in crowded scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 1, pp. 18–32, Jan. 2014.
- [8] C. Lu, J. Shi, W. Wang, and J. Jia, "Fast abnormal event detection," *Int. J. Comput. Vis.*, vol. 127, no. 8, pp. 993–1011, Aug. 2019.
- [9] V. Saligrama and Z. Chen, "Video anomaly detection based on local statistical aggregates," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2112–2119.

- [10] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis, "Learning temporal regularity in video sequences," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 733–742.
- [11] W. Liu, W. Luo, D. Lian, and S. Gao, "Future frame prediction for anomaly detection—A new baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6536–6545.
- [12] W. Sultani, C. Chen, and M. Shah, "Real-world anomaly detection in surveillance videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6479–6488.
- [13] B. Ramachandra, M. J. Jones, and R. R. Vatsavai, "A survey of single-scene video anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2293–2312, May 2022.
- [14] K. K. Santhosh, D. P. Dogra, and P. P. Roy, "Anomaly detection in road traffic using visual surveillance: A survey," *ACM Comput. Surv.*, vol. 53, no. 6, pp. 1–26, Nov. 2021.
- [15] R. Nayak, U. C. Pati, and S. K. Das, "A comprehensive review on deep learning-based methods for video anomaly detection," *Image Vis. Comput.*, vol. 106, Feb. 2021, Art. no. 104078.
- [16] T. M. Tran, T. N. Vu, N. D. Vo, T. V. Nguyen, and K. Nguyen, "Anomaly analysis in images and videos: A comprehensive review," *ACM Comput. Surv.*, vol. 55, no. 7, pp. 1–37, Jul. 2023.
- [17] S. Chandrakala, K. Deepak, and G. Revathy, "Anomaly detection in surveillance videos: A thematic taxonomy of deep models, review and performance analysis," *Artif. Intell. Rev.*, vol. 56, no. 4, pp. 3319–3368, Apr. 2023.
- [18] Y. Liu et al., "Generalized video anomaly event detection: Systematic taxonomy and comparison of deep models," *ACM Comput. Surv.*, vol. 56, no. 7, pp. 1–38, Jul. 2024.
- [19] J. Liu et al., "Networking systems for video anomaly detection: A tutorial and survey," *ACM Comput. Surveys*, vol. 57, no. 10, pp. 1–37, Oct. 2025.
- [20] Y. Liu et al., "Privacy-preserving video anomaly detection: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 3, 2025, doi: 10.1109/TNNLS.2025.3600252.
- [21] B. Ramachandra and M. J. Jones, "Street scene: A new dataset and evaluation protocol for video anomaly detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 2558–2567.
- [22] H. Zhang et al., "Holmes-VAU: Towards long-term video anomaly understanding at any granularity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 13843–13853.
- [23] H. Du et al., "Uncovering what, why and how: A comprehensive benchmark for causation understanding of video anomaly," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18793–18803.
- [24] L. Zhu, Q. Chen, X. Shen, and X. Cun, "VAU-R1: Advancing video anomaly understanding via reinforcement fine-tuning," 2025, *arXiv:2505.23504*.
- [25] R. Nayak, U. C. Pati, and S. K. Das, "A comprehensive review of datasets for detection and localization of video anomalies: A step towards data-centric artificial intelligence-based video anomaly detection," *Multimedia Tools Appl.*, vol. 83, no. 21, pp. 59617–59674, Dec. 2023.
- [26] W. Luo, W. Liu, and S. Gao, "Remembering history with convolutional LSTM for anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2017, pp. 439–444.
- [27] Y. Zhao, B. Deng, C. Shen, Y. Liu, H. Lu, and X.-S. Hua, "Spatio-temporal AutoEncoder for video anomaly detection," in *Proc. 25th ACM Int. Conf. Multimedia*, Oct. 2017, pp. 1933–1941.
- [28] M. Ravanbakhsh, M. Nabi, E. Sangineto, L. Marcenaro, C. Regazzoni, and N. Sebe, "Abnormal event detection in videos using generative adversarial nets," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 1577–1581.
- [29] T. N. Nguyen and J. Meunier, "Anomaly detection in video sequence with appearance-motion correspondence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1273–1283.
- [30] D. Xu, E. Ricci, Y. Yan, J. Song, and N. Sebe, "Learning deep representations of appearance and motion for anomalous event detection," in *Proc. Brit. Mach. Vis. Conf.*, 2015, pp. 8.1–8.12.
- [31] D. Xu, Y. Yan, E. Ricci, and N. Sebe, "Detecting anomalous events in videos by learning deep representations of appearance and motion," *Comput. Vis. Image Understand.*, vol. 156, pp. 117–127, Mar. 2017.
- [32] P. Wu, J. Liu, and F. Shen, "A deep one-class neural network for anomalous event detection in complex scenes," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 7, pp. 2609–2622, Jul. 2020.
- [33] M. Sabokrou, M. Fayyaz, M. Fathy, and R. Klette, "Deep-cascade: Cascading 3D deep neural networks for fast anomaly detection and localization in crowded scenes," *IEEE Trans. Image Process.*, vol. 26, no. 4, pp. 1992–2004, Apr. 2017.
- [34] T. Wang et al., "Generative neural networks for anomaly detection in crowded scenes," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 5, pp. 1390–1399, May 2019.
- [35] Y. Fan, G. Wen, D. Li, S. Qiu, M. D. Levine, and F. Xiao, "Video anomaly detection and localization via Gaussian mixture fully convolutional variational autoencoder," *Comput. Vis. Image Understand.*, vol. 195, Jun. 2020, Art. no. 102920.
- [36] R. Hinami, T. Mei, and S. Satoh, "Joint detection and recounting of abnormal events by learning deep generic knowledge," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3639–3647.
- [37] R. T. Ionescu, F. S. Khan, M.-I. Georgescu, and L. Shao, "Object-centric auto-encoders and dummy anomalies for abnormal event detection in video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7842–7851.
- [38] Z. Liu, Y. Nie, C. Long, Q. Zhang, and G. Li, "A hybrid video anomaly detection framework via memory-augmented flow reconstruction and flow-guided frame prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 13568–13577.
- [39] Q. Bao, F. Liu, Y. Liu, L. Jiao, X. Liu, and L. Li, "Hierarchical scene normality-binding modeling for anomaly detection in surveillance videos," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6103–6112.
- [40] C. Chen et al., "Comprehensive regularization in a bi-directional predictive network for video anomaly detection," in *Proc. AAAI*, 2022, vol. 36, no. 1, pp. 230–238.
- [41] C. Sun, Y. Jia, and Y. Wu, "Evidential reasoning for video anomaly detection," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 2106–2114.
- [42] S. Sun and X. Gong, "Hierarchical semantic contrast for scene-aware video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22846–22856.
- [43] W. Luo, W. Liu, D. Lian, and S. Gao, "Future frame prediction network for video anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7505–7520, Nov. 2022.
- [44] P. Wu, J. Liu, M. Li, Y. Sun, and F. Shen, "Fast sparse coding networks for anomaly detection in videos," *Pattern Recognit.*, vol. 107, Nov. 2020, Art. no. 107515.
- [45] R. Cai, H. Zhang, W. Liu, S. Gao, and Z. Hao, "Appearance-motion memory consistency network for video anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2021, pp. 938–946.
- [46] Y. Liu, J. Liu, X. Zhu, D. Wei, X. Huang, and L. Song, "Learning task-specific representation for video anomaly detection with spatial-temporal attention," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2190–2194.
- [47] X. Huang, C. Zhao, and Z. Wu, "A video anomaly detection framework based on appearance-motion semantics representation consistency," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
- [48] N. Li, F. Chang, and C. Liu, "Spatial-temporal cascade autoencoder for video anomaly detection in crowded scenes," *IEEE Trans. Multimedia*, vol. 23, pp. 203–215, 2021.
- [49] B. Ramachandra, M. J. Jones, and R. R. Vatsavai, "Learning a distance function with a Siamese network to localize anomalies in videos," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Mar. 2020, pp. 2598–2607.
- [50] T. Reiss and Y. Hoshen, "An attribute-based method for video anomaly detection," 2022, *arXiv:2212.00789*.
- [51] R. Morais, V. Le, T. Tran, B. Saha, M. Mansour, and S. Venkatesh, "Learning regularity in skeleton trajectories for anomaly detection in videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 11996–12004.
- [52] A. Markovitz, G. Sharir, I. Friedman, L. Zelnik-Manor, and S. Avidan, "Graph embedded pose clustering for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10539–10547.
- [53] R. Rodrigues, N. Bhargava, R. Velmurugan, and S. Chaudhuri, "Multi-timescale trajectory prediction for abnormal human activity detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 2615–2623.
- [54] W. Luo, W. Liu, and S. Gao, "Normal graph: Spatial temporal graph convolutional networks based prediction network for skeleton based video anomaly detection," *Neurocomputing*, vol. 444, pp. 332–337, Jul. 2021.

- [55] X. Zeng, Y. Jiang, W. Ding, H. Li, Y. Hao, and Z. Qiu, "A hierarchical spatio-temporal graph convolutional neural network for anomaly detection in videos," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 1, pp. 200–212, Jan. 2023.
- [56] Y. Yang, Z. Fu, and S. M. Naqvi, "A two-stream information fusion approach to abnormal event detection in video," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 5787–5791.
- [57] N. Li, F. Chang, and C. Liu, "Human-related anomalous event detection via spatial-temporal graph convolutional autoencoder with embedded long short-term memory network," *Neurocomputing*, vol. 490, pp. 482–494, Jun. 2022.
- [58] C. Huang et al., "Hierarchical graph embedded pose regularity learning via spatio-temporal transformer for abnormal behavior detection," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 307–315.
- [59] O. Hirschorn and S. Avidan, "Normalizing flows for human pose anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 13499–13508.
- [60] S. Yu et al., "Regularity learning via explicit distribution modeling for skeletal video anomaly detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 6661–6673, Aug. 2024.
- [61] A. Flaborea, L. Collorone, G. M. D'Amely Di Melendugno, S. D'Arrigo, B. Prencakj, and F. Galasso, "Multimodal motion conditioned diffusion model for skeleton-based video anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 10284–10295.
- [62] A. Stergiou, B. De Weerd, and N. Deligiannis, "Holistic representation learning for multitask trajectory anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 6715–6725.
- [63] R. Pi, P. Wu, X. He, and Y. Peng, "EOGT: Video anomaly detection with enhanced object information and global temporal dependency," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 20, no. 10, pp. 1–21, Oct. 2024.
- [64] Y. Chang, Z. Tu, W. Xie, and J. Yuan, "Clustering driven deep autoencoder for video anomaly detection," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 329–345.
- [65] Z. Fang, J. Liang, J. T. Zhou, Y. Xiao, and F. Yang, "Anomaly detection with bidirectional consistency in videos," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1079–1092, Mar. 2022.
- [66] C. Huang et al., "Self-supervision-augmented deep autoencoder for unsupervised visual anomaly detection," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13834–13847, Dec. 2022.
- [67] J. T. Zhou, L. Zhang, Z. Fang, J. Du, X. Peng, and Y. Xiao, "Attention-driven loss for anomaly detection in video surveillance," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4639–4647, Dec. 2020.
- [68] Y. Zhang, X. Nie, R. He, M. Chen, and Y. Yin, "Normality learning in multispace for video anomaly detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 9, pp. 3694–3706, Sep. 2021.
- [69] X. Wang et al., "Robust unsupervised video anomaly detection by multipath frame prediction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2301–2312, Jun. 2022.
- [70] J. Yu, Y. Lee, K. C. Yow, M. Jeon, and W. Pedrycz, "Abnormal event detection and localization via adversarial event prediction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 8, pp. 3572–3586, Aug. 2022.
- [71] W. Zhou, Y. Li, and C. Zhao, "Object-guided and motion-refined attention network for video anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.
- [72] K. Cheng, X. Zeng, Y. Liu, M. Zhao, C. Pang, and X. Hu, "Spatial-temporal graph convolutional network boosted flow-frame prediction for video anomaly detection," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
- [73] Y. Liu et al., "AMP-Net: Appearance-motion prototype network assisted automatic video anomaly detection system," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 2843–2855, Feb. 2024.
- [74] G. Yu et al., "Cloze test helps: Effective video anomaly detection via learning to complete video events," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 583–591.
- [75] Z. Yang, J. Liu, Z. Wu, P. Wu, and X. Liu, "Video event restoration based on keyframes for video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14592–14601.
- [76] G. Yu, S. Wang, Z. Cai, X. Liu, E. Zhu, and J. Yin, "Video anomaly detection via visual cloze tests," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4955–4969, 2023.
- [77] G. Wang, Y. Wang, J. Qin, D. Zhang, X. Bao, and D. Huang, "Video anomaly detection by solving decoupled spatio-temporal jigsaw puzzles," in *Proc. ECCV*, 2022, pp. 494–511.
- [78] C. Shi, C. Sun, Y. Wu, and Y. Jia, "Video anomaly detection via sequentially learning multiple pretext tasks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 10296–10306.
- [79] A. Barbalau et al., "SSMTL++: Revisiting self-supervised multi-task learning for video anomaly detection," *Comput. Vis. Image Understand.*, vol. 229, Mar. 2023, Art. no. 103656.
- [80] C. Huang, Z. Wu, J. Wen, Y. Xu, Q. Jiang, and Y. Wang, "Abnormal event detection using deep contrastive learning for intelligent video surveillance system," *IEEE Trans. Ind. Informat.*, vol. 18, no. 8, pp. 5171–5179, Aug. 2022.
- [81] Z. Wang, Y. Zou, and Z. Zhang, "Cluster attention contrast for video anomaly detection," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 2463–2471.
- [82] Y. Lu, C. Cao, Y. Zhang, and Y. Zhang, "Learnable locality-sensitive hashing for video anomaly detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 2, pp. 963–976, Feb. 2023.
- [83] C. Sun, Y. Jia, H. Song, and Y. Wu, "Adversarial 3D convolutional auto-encoder for abnormal event detection in videos," *IEEE Trans. Multimedia*, vol. 23, pp. 3292–3305, 2021.
- [84] D. Chen, L. Yue, X. Chang, M. Xu, and T. Jia, "NM-GAN: Noise-modulated generative adversarial network for video anomaly detection," *Pattern Recognit.*, vol. 116, Aug. 2021, Art. no. 107969.
- [85] W. Luo, W. Liu, and S. Gao, "A revisit of sparse coding based anomaly detection in stacked RNN framework," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 341–349.
- [86] J. T. Zhou, J. Du, H. Zhu, X. Peng, Y. Liu, and R. S. M. Goh, "AnomalyNet: An anomaly detection network for video surveillance," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 10, pp. 2537–2550, Oct. 2019.
- [87] W. Luo et al., "Video anomaly detection with sparse coding inspired deep neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 3, pp. 1070–1084, Mar. 2021.
- [88] V. Zavrtnik, M. Kristan, and D. Škočaj, "Reconstruction by inpainting for visual anomaly detection," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107706.
- [89] N.-C. Ristea et al., "Self-supervised predictive convolutional attentive block for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13576–13586.
- [90] N. Madan et al., "Self-supervised masked convolutional transformer block for anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 525–542, Jan. 2024.
- [91] N.-C. Ristea, F.-A. Croitoru, R. T. Ionescu, M. Popescu, F. S. Khan, and M. Shah, "Self-distilled masked auto-encoders are efficient video anomaly detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15984–15995.
- [92] M. Ye, X. Peng, W. Gan, W. Wu, and Y. Qiao, "AnoPCN: Video anomaly detection via deep predictive coding network," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1805–1813.
- [93] Y. Liu, J. Liu, J. Lin, M. Zhao, and L. Song, "Appearance-motion united auto-encoder framework for video anomaly detection," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 5, pp. 2498–2502, May 2022.
- [94] Y. Liu, J. Liu, M. Zhao, D. Yang, X. Zhu, and L. Song, "Learning appearance-motion normality for video anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.
- [95] C. Huang et al., "Self-supervised attentive generative adversarial networks for video anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9389–9403, Nov. 2023.
- [96] M.-I. Georgescu, A. Barbalau, R. T. Ionescu, F. S. Khan, M. Popescu, and M. Shah, "Anomaly detection in video via self-supervised and multi-task learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 12742–12752.
- [97] M. Zhang et al., "Multi-scale video anomaly detection by multi-grained spatio-temporal representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17385–17394.
- [98] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.
- [99] D. M. J. Tax and R. P. W. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, 2003.
- [100] J. Wang and A. Cherian, "GODS: Generalized one-class discriminative subspaces for anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8200–8210.

- [101] L. Ruff et al., "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4393–4402.
- [102] P. Liznerski, L. Ruff, R. A. Vandermeulen, B. J. Franks, M. Kloft, and K. Müller, "Explainable deep one-class classification," in *Proc. ICLR*, 2020, pp. 1–25.
- [103] M. Sabokrou, M. Fayyaz, M. Fathy, Z. Moayed, and R. Klette, "Deep-anomaly: Fully convolutional neural network for fast anomaly detection in crowded scenes," *Comput. Vis. Image Understand.*, vol. 172, pp. 88–97, Jul. 2018.
- [104] M. Sabokrou, M. Khalooei, M. Fathy, and E. Adeli, "Adversarially learned one-class classifier for novelty detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3379–3388.
- [105] M. Sabokrou et al., "AVID: Adversarial visual irregularity detection," in *Proc. Asian Conf. Comput. Vis.*, 2018, pp. 488–505.
- [106] M. Sabokrou, M. Fathy, G. Zhao, and E. Adeli, "Deep end-to-end one-class classifier," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 2, pp. 675–684, Feb. 2021.
- [107] M. Zaigham Zaheer, J.-H. Lee, M. Astrid, and S.-I. Lee, "Old is gold: Redefining the adversarially learned one-class classifier training paradigm," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14171–14181.
- [108] M. Z. Zaheer, J.-H. Lee, A. Mahmood, M. Astrid, and S.-I. Lee, "Stabilizing adversarially learned one-class novelty detection using pseudo anomalies," *IEEE Trans. Image Process.*, vol. 31, pp. 5963–5975, 2022.
- [109] K. Doshi and Y. Yilmaz, "Towards interpretable video anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 2654–2663.
- [110] A. Singh, M. J. Jones, and E. G. Learned-Miller, "EVAL: Explainable video anomaly localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18717–18726.
- [111] Y. Yang, K. Lee, B. Dariush, Y. Cao, and S.-Y. Lo, "Follow the rules: Reasoning for video anomaly detection with large language models," in *Proc. ECCV*. Cham, Switzerland: Springer, 2024, pp. 304–322.
- [112] C. Li and Y. Jiang, "VLAVAD: Vision-language models assisted unsupervised video anomaly detection," in *Proc. BMVC*, 2024, pp. 1–15.
- [113] Z. Ding et al., "SlowFastVAD: Video anomaly detection via integrating simple detector and RAG-enhanced vision-language model," 2025, *arXiv:2504.10320*.
- [114] D. Gong et al., "Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1705–1714.
- [115] S. Lee, H. G. Kim, and Y. M. Ro, "BMAN: Bidirectional multi-scale aggregation networks for abnormal event detection," *IEEE Trans. Image Process.*, vol. 29, pp. 2395–2408, 2020.
- [116] H. Park, J. Noh, and B. Ham, "Learning memory-guided normality for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14372–14381.
- [117] H. Lv, C. Chen, Z. Cui, C. Xu, Y. Li, and J. Yang, "Learning normal dynamics in videos with meta prototype network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 15425–15434.
- [118] M. I. Georgescu, R. Ionescu, F. N. Khan, M. Popescu, and M. Shah, "A background-agnostic framework with adversarial training for abnormal event detection in video," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4505–4523, Sep. 2021.
- [119] Z. Yang, P. Wu, J. Liu, and X. Liu, "Dynamic local aggregation network with adaptive clusterer for anomaly detection," in *Proc. ECCV*, 2022, pp. 404–421.
- [120] G. Yu, S. Wang, Z. Cai, X. Liu, and C. Wu, "Effective video abnormal event detection by learning a consistency-aware high-level feature extractor," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6337–6346.
- [121] C. Sun, C. Shi, Y. Jia, and Y. Wu, "Learning event-relevant factors for video anomaly detection," in *Proc. AAAI*, 2023, vol. 37, no. 2, pp. 2384–2392.
- [122] W. Liu, H. Chang, B. Ma, S. Shan, and X. Chen, "Diversity-measurable anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12147–12156.
- [123] C. Cao, Y. Lu, P. Wang, and Y. Zhang, "A new comprehensive benchmark for semi-supervised video anomaly detection and anticipation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20392–20401.
- [124] C. Yan, S. Zhang, Y. Liu, G. Pang, and W. Wang, "Feature prediction diffusion model for video anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 5527–5537.
- [125] Z. Liu, X.-M. Wu, D. Zheng, K.-Y. Lin, and W.-S. Zheng, "Generating anomalies for video anomaly detection with prompt-based feature mapping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24500–24510.
- [126] C. Cao, H. Zhang, Y. Lu, P. Wang, and Y. Zhang, "Scene-dependent prediction in latent space for video anomaly detection and anticipation," in *Proc. IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, Jan. 2024, pp. 224–239.
- [127] H. Hu, W. Du, P. Liao, B. Wang, and S. Fan, "Noise-resistant video anomaly detection via RGB error-guided multiscale predictive coding and dynamic memory," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 19109–19119.
- [128] J. Ryan Medel and A. Savakis, "Anomaly detection in video using predictive convolutional long short-term memory networks," 2016, *arXiv:1612.00390*.
- [129] M. Ravanbakhsh, E. Sangineto, M. Nabi, and N. Sebe, "Training adversarial discriminators for cross-channel abnormal event detection in crowds," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1896–1904.
- [130] H. Vu, T. D. Nguyen, T. Le, W. Luo, and D. Phung, "Robust anomaly detection in videos using multilevel representations," in *Proc. AAAI*, 2019, vol. 33, no. 1, pp. 5216–5223.
- [131] H. Song, C. Sun, X. Wu, M. Chen, and Y. Jia, "Learning normal patterns via adversarial attention-based autoencoder for abnormal event detection in videos," *IEEE Trans. Multimedia*, vol. 22, no. 8, pp. 2138–2148, Aug. 2020.
- [132] X. Feng, D. Song, Y. Chen, Z. Chen, J. Ni, and H.-F. Chen, "Convolutional transformer based dual discriminator generative adversarial networks for video anomaly detection," in *Proc. ACM MM*, 2021, pp. 5546–5554.
- [133] H. Zhou et al., "Video anomaly detection with motion and appearance guided patch diffusion model," in *Proc. AAAI*, 2025, vol. 39, no. 10, pp. 10761–10769.
- [134] P. Wu, W. Wang, F. Chang, C. Liu, and B. Wang, "DSS-Net: Dynamic self-supervised network for video anomaly detection," *IEEE Trans. Multimedia*, vol. 26, pp. 2124–2136, 2024.
- [135] M. Astrid, M. Z. Zaheer, and S.-I. Lee, "Limiting reconstruction capability of autoencoders using moving backward pseudo anomalies," in *Proc. 19th Int. Conf. Ubiquitous Robots (UR)*, Jul. 2022, pp. 248–251.
- [136] M. Astrid, M. Z. Zaheer, J.-Y. Lee, and S.-I. Lee, "Learning not to reconstruct anomalies," in *Proc. Brit. Mach. Vis. Conf.*, 2021.
- [137] M. Astrid, M. Z. Zaheer, and S.-I. Lee, "PseudoBound: Limiting the anomaly reconstruction capability of one-class classifiers using pseudo anomalies," *Neurocomputing*, vol. 534, pp. 147–160, May 2023.
- [138] M. Pourreza, B. Mohammadi, M. Khaki, S. Bouindour, H. Snoussi, and M. Sabokrou, "G2D: Generate to detect anomaly," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 2003–2012.
- [139] R. Delussu, L. Putzu, and G. Fumera, "Synthetic data for video surveillance applications of computer vision: A review," *Int. J. Comput. Vis.*, vol. 132, no. 10, pp. 4473–4509, Oct. 2024.
- [140] J. Leng, M. Tan, X. Gao, W. Lu, and Z. Xu, "Anomaly warning: Learning and memorizing future semantic patterns for unsupervised ex-ante potential anomaly prediction," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6746–6754.
- [141] Y. Liu et al., "Stochastic video normality network for abnormal event detection in surveillance videos," *Knowl.-Based Syst.*, vol. 280, Nov. 2023, Art. no. 110986.
- [142] L. Wang, J. Tian, S. Zhou, H. Shi, and G. Hua, "Memory-augmented appearance-motion network for video anomaly detection," *Pattern Recognit.*, vol. 138, Jun. 2023, Art. no. 109335.
- [143] C. Cao, Y. Lu, and Y. Zhang, "Context recovery and knowledge retrieval: A novel two-stream framework for video anomaly detection," *IEEE Trans. Image Process.*, vol. 33, pp. 1810–1825, 2024.
- [144] C. Huang et al., "Pixel-level anomaly detection via uncertainty-aware prototypical transformer," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 521–530.
- [145] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 4489–4497.
- [146] M. Z. Zaheer, A. Mahmood, M. Astrid, and S.-I. Lee, "CLAWS: Clustering assisted weakly supervised learning with normalcy suppression for anomalous event detection," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 358–376.
- [147] J. Carreira and A. Zisserman, "Quo vadis, action recognition? A new model and the kinetics dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6299–6308.

- [148] P. Wu et al., "Not only look, but also listen: Learning multimodal violence detection under weak supervision," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 322–339.
- [149] J.-C. Wu, H.-Y. Hsieh, D.-J. Chen, C.-S. Fuh, and T.-L. Liu, "Self-supervised sparse representation for video anomaly detection," in *Proc. 17th Eur. Conf. Comput. Vis.*, 2022, pp. 729–745.
- [150] Y. Zhou, Y. Qu, X. Xu, F. Shen, J. Song, and H. Shen, "BatchNorm-based weakly supervised video anomaly detection," 2023, *arXiv:2311.15367*.
- [151] S. AlMarri, M. Z. Zaheer, and K. Nandakumar, "A multi-head approach with shuffled segments for weakly-supervised video anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2024, pp. 132–142.
- [152] K. Hara, H. Kataoka, and Y. Satoh, "Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet?," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6546–6555.
- [153] S. Sun and X. Gong, "Long-short temporal co-teaching for weakly supervised video anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 2711–2716.
- [154] L. Wang et al., "Temporal segment networks: Towards good practices for deep action recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 20–36.
- [155] J.-X. Zhong, N. Li, W. Kong, S. Liu, T. H. Li, and G. Li, "Graph convolutional label noise cleaner: Train a plug-and-play action classifier for anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1237–1246.
- [156] N. Li, J.-X. Zhong, X. Shu, and H. Guo, "Weakly-supervised anomaly detection in video surveillance via graph convolutional label noise cleaning," *Neurocomputing*, vol. 481, pp. 154–167, Apr. 2022.
- [157] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
- [158] S. Li, F. Liu, and L. Jiao, "Self-training multi-sequence learning with transformer for weakly supervised video anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 2, 2022, pp. 1395–1403.
- [159] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
- [160] P. Wu et al., "Open-vocabulary video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18297–18307.
- [161] B. Wan, Y. Fang, X. Xia, and J. Mei, "Weakly supervised video anomaly detection via center-guided discriminative learning," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2020, pp. 1–6.
- [162] S. Hershey et al., "CNN architectures for large-scale audio classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 131–135.
- [163] W.-F. Pang, Q.-H. He, Y.-J. Hu, and Y.-X. Li, "Violence detection in videos based on fusing visual and audio information," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 2260–2264.
- [164] X. Peng et al., "Learning weakly supervised audio-visual violence detection in hyperbolic space," 2023, *arXiv:2305.18797*.
- [165] Y. Pu, X. Wu, L. Yang, and S. Wang, "Learning prompt-enhanced context features for weakly-supervised video anomaly detection," *IEEE Trans. Image Process.*, vol. 33, pp. 4923–4936, 2024.
- [166] W. Chen, K. T. Ma, Z. Jian Yew, M. Hur, and D. A.-A. Khoo, "TEVAD: Improved video anomaly detection with captions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 5549–5559.
- [167] C. Tao et al., "Learn suspected anomalies from event prompts for video anomaly detection," 2024, *arXiv:2403.01169*.
- [168] P. Wu, J. Liu, X. He, Y. Peng, P. Wang, and Y. Zhang, "Toward video anomaly retrieval from video anomaly detection: New benchmarks and model," *IEEE Trans. Image Process.*, vol. 33, pp. 2213–2225, 2024.
- [169] D.-L. Wei, C.-G. Liu, Y. Liu, J. Liu, X.-G. Zhu, and X.-H. Zeng, "Look, listen and pay more attention: Fusing multi-modal information for video violence detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 1980–1984.
- [170] D. Wei, Y. Liu, X. Zhu, J. Liu, and X. Zeng, "MSAF: Multimodal supervise-attention enhanced fusion for video anomaly detection," *IEEE Signal Process. Lett.*, vol. 29, pp. 2178–2182, 2022.
- [171] J. Yu, J. Liu, Y. Cheng, R. Feng, and Y. Zhang, "Modality-aware contrastive instance learning with self-distillation for weakly-supervised audio-visual violence detection," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6278–6287.
- [172] P. Wu, X. Liu, and J. Liu, "Weakly supervised audio-visual violence detection," *IEEE Trans. Multimedia*, vol. 25, pp. 1674–1685, 2023.
- [173] T. Yuan et al., "Towards surveillance video-and-language understanding: New dataset, baselines, and challenges," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 22052–22061.
- [174] Y. Zhu and S. Newsam, "Motion-aware feature for improved video anomaly detection," in *Proc. BMVC*, 2019, pp. 270–281.
- [175] J. Zhang, L. Qing, and J. Miao, "Temporal convolutional network with complementary inner bag loss for weakly supervised anomaly detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 4030–4034.
- [176] Y. Liu, J. Liu, M. Zhao, S. Li, and L. Song, "Collaborative normality learning framework for weakly supervised video anomaly detection," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 5, pp. 2508–2512, May 2022.
- [177] J.-C. Feng, F.-T. Hong, and W.-S. Zheng, "MIST: Multiple instance self-training framework for video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14009–14018.
- [178] C. Zhang et al., "Exploiting completeness and uncertainty of pseudo labels for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 16271–16280.
- [179] Z. Yang, J. Liu, and P. Wu, "Text prompt with normality guidance for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18899–18908.
- [180] P. Wu et al., "VadCLIP: Adapting vision-language models for weakly supervised video anomaly detection," in *Proc. AAAI*, 2024, vol. 38, no. 6, pp. 6074–6082.
- [181] Q. Chen et al., "HAWK: Learning to understand open-world video anomalies," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 139751–139785.
- [182] M. Ye, W. Liu, and P. He, "VERA: Explainable video anomaly detection via verbalized learning of vision-language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 8679–8688.
- [183] H. Lv and Q. Sun, "Video anomaly detection and explanation via large language models," 2024, *arXiv:2401.05702*.
- [184] L. Zanella, W. Menapace, M. Mancini, Y. Wang, and E. Ricci, "Harnessing large language models for training-free video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18527–18536.
- [185] Y. Tian, G. Pang, Y. Chen, R. Singh, J. W. Verjans, and G. Carneiro, "Weakly-supervised video anomaly detection with robust temporal feature magnitude learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4975–4986.
- [186] P. Wu and J. Liu, "Learning causal temporal relation and feature discrimination for anomaly detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3513–3527, 2021.
- [187] G. Li, G. Cai, X. Zeng, and R. Zhao, "Scale-aware spatio-temporal relation learning for video anomaly detection," in *Proc. ECCV*, 2022, pp. 333–350.
- [188] M. Cho, M. Kim, S. Hwang, C. Park, K. Lee, and S. Lee, "Look around for anomalies: Weakly-supervised anomaly detection via context-motion relational learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12137–12146.
- [189] Y. Chen, Z. Liu, B. Zhang, W. Fok, X. Qi, and Y.-C. Wu, "MGFN: Magnitude-contrastive glance-and-focus network for weakly-supervised video anomaly detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 1, pp. 387–395.
- [190] H. Lv, Z. Yue, Q. Sun, B. Luo, Z. Cui, and H. Zhang, "Unbiased multiple instance learning for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 8022–8031.
- [191] H. Zhou, J. Yu, and W. Yang, "Dual memory units with uncertainty regulation for weakly supervised video anomaly detection," in *Proc. AAAI*, 2023, vol. 37, no. 3, pp. 3769–3777.
- [192] J. Chen, L. Li, L. Su, Z.-J. Zha, and Q. Huang, "Prompt-enhanced multiple instance learning for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18319–18329.
- [193] P. Wu et al., "Weakly supervised video anomaly detection and localization with spatio-temporal prompts," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 9301–9310.

- [194] T. Liu, C. Zhang, K.-M. Lam, and J. Kong, "Decouple and resolve: Transformer-based models for online anomaly detection from weakly labeled videos," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 15–28, 2023.
- [195] S. Chang, Y. Li, S. Shen, J. Feng, and Z. Zhou, "Contrastive attention for video anomaly detection," *IEEE Trans. Multimedia*, vol. 24, pp. 4067–4076, 2022.
- [196] D. Purwanto, Y.-T. Chen, and W.-H. Fang, "Dance with self-attention: A new look of conditional random fields on anomaly detection in videos," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 173–183.
- [197] C. Huang et al., "Weakly supervised video anomaly detection via self-guided temporal discriminative transformer," *IEEE Trans. Cybern.*, vol. 54, no. 5, pp. 3197–3210, May 2024.
- [198] C. Zhang, G. Li, Q. Xu, X. Zhang, L. Su, and Q. Huang, "Weakly supervised anomaly detection in videos considering the openness of events," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21687–21699, Nov. 2022.
- [199] C. Zhang et al., "Dynamic erasing network with adaptive temporal modeling for weakly supervised video anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 16706–16720, Sep. 2025.
- [200] H. Ye, K. Xu, X. Jiang, and T. Sun, "Learning spatio-temporal relations with multi-scale integrated perception for video anomaly detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 4020–4024.
- [201] S. Lin, H. Yang, X. Tang, T. Shi, and L. Chen, "Social MIL: Interaction-aware for crowd anomaly detection," in *Proc. 16th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2019, pp. 1–8.
- [202] S. Park, H. Kim, M. Kim, D. Kim, and K. Sohn, "Normality guided multiple instance learning for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 2664–2673.
- [203] J. Gong, C. Wang, X. Dai, S. Yu, L. Xiang, and J. Wu, "Multi-scale continuity-aware refinement network for weakly supervised video anomaly detection," in *Proc. IEEE Int. Conf. Multimedia Expo. (ICME)*, Jul. 2022, pp. 1–6.
- [204] H. Sapkota and Q. Yu, "Bayesian nonparametric submodular video partition for robust anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 3212–3221.
- [205] J. Fioresi, I. R. Dave, and M. Shah, "TeD-SPAD: Temporal distinctiveness for self-supervised privacy-preservation for video anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 13552–13563.
- [206] M. Z. Zaheer, A. Mahmood, M. Astrid, and S.-I. Lee, "Clustering aided weakly supervised training to detect anomalous events in surveillance videos," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14085–14098, Oct. 2024.
- [207] T. Liu, K.-M. Lam, and J. Kong, "Distilling privileged knowledge for anomalous event detection from weakly labeled videos," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 1–15, Sep. 2024.
- [208] H. K. Joo, K. Vo, K. Yamazaki, and N. Le, "CLIP-TSA: Clip-assisted temporal self-attention for weakly-supervised video anomaly detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2023, pp. 3230–3234.
- [209] K. Liu and H. Ma, "Exploring background-bias for anomaly detection in surveillance videos," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1490–1499.
- [210] J. Wu et al., "Weakly-supervised spatio-temporal anomaly detection in surveillance video," in *Proc. Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 1172–1178.
- [211] Z. Dong, J. Qin, and Y. Wang, "Multi-stream deep networks for person to person violence detection in videos," in *Proc. 7th Chin. Conf. Pattern Recognit. (CCPR)*, Nov. 2016, pp. 517–531.
- [212] P. Zhou, Q. Ding, H. Luo, and X. Hou, "Violent interaction detection in video based on deep learning," *J. Phys., Conf. Ser.*, vol. 844, Jun. 2017, Art. no. 012044.
- [213] S. Sudhakaran and O. Lanz, "Learning to detect violent videos using convolutional long short-term memory," in *Proc. 14th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2017, pp. 1–6.
- [214] A. Hanson, K. Pnvr, S. Krishnagopal, and L. S. Davis, "Bidirectional convolutional LSTM for the detection of violence in videos," in *Proc. ECCV Workshop*, 2019, pp. 280–295.
- [215] Y. Su, G. Lin, J. Zhu, and Q. Wu, "Human interaction learning on 3D skeleton point clouds for video violence recognition," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 74–90.
- [216] M. Cheng, K. Cai, and M. Li, "RWF-2000: An open large scale video database for violence detection," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 4183–4190.
- [217] J. Su, P. Her, E. Clemens, E. Yaz, S. Schneider, and H. Medeiros, "Violence detection using 3D convolutional neural networks," in *Proc. 18th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Nov. 2022, pp. 1–8.
- [218] G. Garcia-Cobo and J. C. SanMiguel, "Human skeletons and change detection for efficient violence detection in surveillance videos," *Comput. Vis. Image Understand.*, vol. 233, Aug. 2023, Art. no. 103739.
- [219] P. Zhang, X. Zhao, L. Dong, W. Lei, W. Zhang, and Z. Lin, "A framework for detecting fighting behavior based on key points of human skeletal posture," *Comput. Vis. Image Understand.*, vol. 248, Nov. 2024, Art. no. 104123.
- [220] B. Peixoto, B. Lavi, J. P. P. Martin, S. Avila, Z. Dias, and A. Rocha, "Toward subjective violence detection in videos," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 8276–8280.
- [221] M. Perez, A. C. Kot, and A. Rocha, "Detection of real-world fights in surveillance videos," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 2662–2666.
- [222] B. Peixoto, B. Lavi, P. Bestagini, Z. Dias, and A. Rocha, "Multimodal violence detection in videos," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2957–2961.
- [223] A. Singh, D. Patil, and S. N. Omkar, "Eye in the sky: Real-time drone surveillance system (DSS) for violent individuals identification using ScatterNet hybrid deep learning network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 1710–17108.
- [224] Y. Shang, X. Wu, and R. Liu, "Multimodal violent video recognition based on mutual distillation," in *Proc. PRCV*, 2022, pp. 623–637.
- [225] A. Del Giorno, J. A. Bagnell, and M. Hebert, "A discriminative framework for anomaly detection in large videos," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 334–349.
- [226] R. T. Ionescu, S. Smeureanu, B. Alexe, and M. Popescu, "Unmasking the abnormal events in video," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2895–2903.
- [227] Y. Liu, C. Li, and B. Póczos, "Classifier two sample test for video anomaly detections," in *Proc. BMVC*, 2018, p. 71.
- [228] S. Wang, Y. Zeng, Q. Liu, C. Zhu, E. Zhu, and J. Yin, "Detecting abnormality without knowing normality: A two-stage approach for unsupervised video abnormal event detection," in *Proc. 26th ACM Int. Conf. Multimedia*, Oct. 2018, pp. 636–644.
- [229] G. Pang, C. Yan, C. Shen, A. Van Den Hengel, and X. Bai, "Self-trained deep ordinal regression for end-to-end video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12173–12182.
- [230] J. Hu, G. Yu, S. Wang, E. Zhu, Z. Cai, and X. Zhu, "Detecting anomalous events from unlabeled videos via temporal masked auto-encoding," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.
- [231] X. Lin, Y. Chen, G. Li, and Y. Yu, "A causal inference look at unsupervised video anomaly detection," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 1620–1629.
- [232] G. Yu, S. Wang, Z. Cai, X. Liu, C. Xu, and C. Wu, "Deep anomaly discovery from unlabeled videos via normality advantage and self-paced refinement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13987–13998.
- [233] H. Shi, L. Wang, S. Zhou, G. Hua, and W. Tang, "Learning anomalies with normality prior for unsupervised video anomaly detection," in *Proc. ECCV*. Cham, Switzerland: Springer, 2024, pp. 163–180.
- [234] C. Tao, C. Wang, S. Lin, S. Cai, D. Li, and J. Qian, "Feature reconstruction with disruption for unsupervised video anomaly detection," *IEEE Trans. Multimedia*, vol. 26, pp. 10160–10173, 2024.
- [235] A. Al-Lahham, M. Z. Zaheer, N. Tasthan, and K. Nandakumar, "Collaborative learning of anomalies with privacy (CLAP) for unsupervised video anomaly detection: A new baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 12416–12425.
- [236] M. Z. Zaheer, A. Mahmood, M. H. Khan, M. Segu, F. Yu, and S.-I. Lee, "Generative cooperative learning for unsupervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14744–14754.
- [237] A. Al-Lahham, N. Tasthan, M. Z. Zaheer, and K. Nandakumar, "A coarse-to-fine pseudo-labeling (C2FPL) framework for unsupervised video anomaly detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 6779–6788.

- [238] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
- [239] T. Li, Z. Wang, S. Liu, and W.-Y. Lin, "Deep unsupervised anomaly detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3635–3644.
- [240] W. Liu, W. Luo, Z. Li, P. Zhao, and S. Gao, "Margin learning embedded prediction for video anomaly detection with a few anomalies," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 3023–3030.
- [241] A. Acisintoe et al., "UBnormal: New benchmark for supervised open-set video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20143–20153.
- [242] Y. Zhu, W. Bao, and Q. Yu, "Towards open set video anomaly detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2022, pp. 395–412.
- [243] C. Ding, G. Pang, and C. Shen, "Catching both gray and black swans: Open-set supervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7388–7398.
- [244] J. Zhu, C. Ding, Y. Tian, and G. Pang, "Anomaly heterogeneity learning for open-set supervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17616–17626.
- [245] Y. Lu, F. Yu, M. K. K. Reddy, and Y. Wang, "Few-shot scene-adaptive anomaly detection," in *Proc. Eur. Conf. Comput. Vis.*, vol. 12350, 2020, pp. 125–141.
- [246] Y. Hu, X. Huang, and X. Luo, "Adaptive anomaly detection network for unseen scene without fine-tuning," in *Proc. PRCV*, 2021, pp. 311–323.
- [247] X. Huang, Y. Hu, X. Luo, J. Han, B. Zhang, and X. Cao, "Boosting variational inference with margin learning for few-shot scene-adaptive anomaly detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 6, pp. 2813–2825, Jun. 2023.
- [248] A. Aich, K.-C. Peng, and A. K. Roy-Chowdhury, "Cross-domain video anomaly detection without target domain adaptation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 2579–2591.
- [249] Y. Jain, A. Dabouei, and M. Xu, "Cross-domain learning for video anomaly detection with limited supervision," in *Proc. ECCV*, 2024, pp. 468–484.
- [250] Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, "AnomalyCLIP: Object-agnostic prompt learning for zero-shot anomaly detection," in *Proc. ICLR*, 2023, pp. 1–33.



Peng Wu received the B.E. and Ph.D. degrees from Xidian University, Xi'an, China, in 2017 and 2022, respectively.

He is currently an Associate Professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an. He has authored over 40 papers in the top-tier journals and conferences, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), European Conference on Computer Vision (ECCV), ACM International Conference on Multimedia (ACM MM), and AAAI Conference on Artificial Intelligence (AAAI). His current research interests include video anomaly detection and vision-language models.



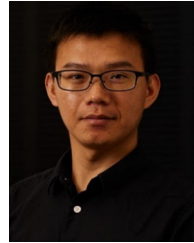
Chengyu Pan received the B.S. degree from Jilin University, Changchun, China, in 2023, where he is currently pursuing the master's degree in computer science and technology with Northwestern Polytechnical University, Xi'an, China.

His current research interests include video anomaly detection and deep learning.



Yuting Yan received the B.E. degree from Northwestern Polytechnical University, Xi'an, China, in 2025, where he is currently pursuing the master's degree in computer science and technology.

His current research interests include event-based vision and multimodal video anomaly detection.



Guansong Pang (Member, IEEE) received the Ph.D. degree from the University of Technology Sydney, Sydney, NSW, Australia, in 2019.

He was a Research Fellow with Australian Institute for Machine Learning, The University of Adelaide, Adelaide, Australia. He is currently a tenure-track Assistant Professor with the School of Computing and Information Systems, Singapore Management University, Singapore. His research focuses on recognizing and generalizing to abnormal/unknown/unseen data for learning trustworthy artificial intelligence (AI) systems in open environments.



Qingsen Yan (Member, IEEE) received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2019.

He was a joint-training Ph.D. Student with The University of Adelaide in 2018, where he was a Research Fellow with Australian Institute for Machine Learning (AIML). He is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. His research interests include high dynamic range imaging, deep learning, medical image analysis, and computer vision.



Peng Wang (Member, IEEE) received the B.E. degree in electrical engineering and automation and the Ph.D. degree in control science and engineering from Beihang University, Beijing, China, in 2004 and 2011, respectively.

He is currently a Professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an, China. He has authored/co-authored more than 100 papers in top-tier journals and conferences, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, *International Journal of Computer Vision*, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), IEEE International Conference on Computer Vision (ICCV), European Conference on Computer Vision (ECCV), and AAAI Conference on Artificial Intelligence (AAAI). His current research interests include computer vision, machine learning, and artificial intelligence.



Yanning Zhang (Fellow, IEEE) received the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 1993 and 1996, respectively.

She is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. She has authored/co-authored more than 200 papers in international journals, conferences, and Chinese key journals. Her research interests include signal and image processing, computer vision, and pattern recognition.